

Future Energy Systems Analysis at Huseby Bruk

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Knowledge Innovation Community

Future Energy Systems Analysis at Huseby Bruk

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Abstract

The objective of this project is to propose a set of changes to Huseby Bruk's energy system to pursue the goal of being energy self-sufficient to increase resiliency, decrease costs, attract visitors, and continue the property's legacy of energy innovation and independence. The main topics of investigation are modifying the electrical grid structure and increasing local renewable energy production. The goal is to design potential solutions and integrate them to meet Huseby Bruk's goals. To approach the problem, previous work done by Sweco on these topics was reviewed. While there is no budget for this project, the task is to show that these changes would be worth the investment and therefore be a valuable and worthwhile project for Sweden's National Property Board to pursue.

Huseby Bruk is a historic property in southern Sweden that welcomes tourists to explore what life was like in the past centuries. In the early 20th century, a hydroelectric plant was built and Huseby Bruk became one of the first properties in its area to produce electricity. As a historic property owned by the National Property Board, Huseby Bruk has certain limitations on what changes can be made to the property.

The property currently generates energy from a hydroelectric plant and a solar photovoltaic (PV) system on one rooftop. The energy production from both sources is seasonal and intermittent due to weather and maintenance issues. The manor house on the property is the biggest energy consumer, consuming more than a quarter of Huseby Bruk's yearly energy consumption.

Increasing renewable energy generation would enhance resiliency, attract visitors, and further Huseby Bruk's legacy as an energy leader. Four types of energy generation were explored: solar, wind, bioenergy, and hydropower. Energy storage options were also considered. For each potential solution, resource availability, production capacity, cost, and feasibility were considered.

The final proposed solution includes a solar PV carport and small-scale vertical axis wind turbines located in the parking lot of Huseby Bruk, as well as a biogas production facility located on the farm. Ultimately, the recommendation is for Huseby Bruk to develop a microgrid that can operate fully independently from the regional electrical grid to maximize resilience and self-sufficiency and decrease energy costs. Huseby Bruk was an energy innovator when it was the first property in the area to electrify, and it can continue its legacy of energy leadership and resilience by becoming energy self-sufficient through the design of a microgrid powered by local renewable energy production.

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Nomenclature

Variable/Parameter	Symbol	Units
Mass	m	kg
Energy	E	kWh
Electrical energy	E	kWh(e)
Volume	V	m ³
Specific volume	v	L/kg
Lower Heating Value (LHV)	LHV	MWh/ton
Digestor volume	V _{DG}	m ³
Money	\$	SEK
Voltage	U	V
Current	I	A
Irradiance	G	W/m ²
Power	W	J/s
Angle	∠	°
Radiation	H	kWh/m ²
Speed	v	m/s
Peak Sun Hour	PSH	kWh/m ²

Acronyms

<u>Acronym</u>	<u>Definition</u>
AC	Alternating current
BAU	Business as usual
BESS	Battery energy storage system
CAPEX	Capital expenditure
CEP	Clean Energy for all Europeans Package
CHP	Combined heat and power
CO ₂	Carbon dioxide
DC	Direct current
ESS	Energy storage system
EU	European Union
GM	Ground mounted
HAWT	Horizontal-axis wind turbine
HRT	Hydraulic retention time
H ₂ S	Hydrogen sulfide
IEA	International Energy Agency
ICE	Internal Combustion Engine
IKN	Non-concessioned networks
IRR	Internal rate of return
KTH	Kungliga Tekniska Högskolan
LCOS	Levelized Cost of Storage
LHV	Lower heating value
MPPT	Maximum power point tracking
NREL	National Renewable Energy Laboratory
OPEX	Operating expenditure
ORC	Organic Rankine cycle
PSH	Peak sun hours
PV	Photovoltaic
SEK	Swedish Krona
SFV	<i>Statens fastighetsverk</i> (National Property Board)
STEPS	Stated Policies Scenario
TSO	Transmission system operator
UPC	Universitat Politècnica de Catalunya - Barcelona Tech
USD	United States dollar
VAT	Value added tax
VAWT	Vertical-axis wind turbine
VPP	Virtual power plant
W _p	Watt-peak
PSH	Peak Sun Hour

1 Introduction

1.1 Introduction and Background of the Project

Huseby Bruk is a historic property in southern Sweden that has a long history of industry, farming, and energy production. Between the 17th and 19th centuries, Huseby Bruk's economy was based on bog iron, produced using materials found on the property: iron ore from the lakes, charcoal from the forests, and lime from the bones of farm animals. The Helge River that flows through the property provided mechanical power to Huseby Bruk in many forms throughout the centuries. In 1905, Huseby Bruk harnessed this hydropower for electricity generation and became the first property in the area to be electrified. The property produced so much electricity that they offered to power the nearby town. Huseby Bruk was bequeathed to the Swedish state in 1979 and is currently owned by the National Property Board (*Statens Fastighetsverk (SFV)*). Huseby Bruk is the most visited tourist destination in the county, with 150,000 visitors each year.

The goal of this project is to continue Huseby Bruk's legacy of energy innovation and independence by increasing resiliency, decreasing energy costs, and attracting visitors to Huseby Bruk.



Figure 1.1. Map of Huseby Bruk

Huseby Bruk contains many historical buildings including a manor house, forge, mill, general store, sawmill, carpentry, stable, and dairy (Figure 1.1). Many forms of economic and industrial activity have sustained the property throughout the centuries, starting in the early 1600s. Iron production was a major industry at Huseby Bruk for over 300 years as everything needed to produce iron was found on the property: bog-iron ore from the lakes, charcoal from the forests (used for the firing of furnaces) and the river that provided power and a means for transportation (Huseby Bruk, 2016). In 1844, the manor house (“*slottet*”) was built (Figure 1.2). In 1905, a hydroelectric plant was built and Huseby Bruk became one of the first properties in Sweden to be electrified. The plant produced so much electricity that Huseby Bruk offered to power the nearby town. Today, there is a hydroelectric plant which is now home to a small electricity museum featuring old switchgear and

a selection of historical electrical equipment. Each year, tens of thousands of tourists come to Huseby Bruk to enjoy the popular Christmas and Easter markets.



Figure 1.2. Manor house ("slottet") at Huseby Bruk

Huseby Bruk is interested in making changes to their energy system to decrease energy costs and increase energy resiliency.

Energy costs are high at Huseby Bruk. Each building has at least one connection point to the regional electricity grid which costs the property a lot of money in grid subscription fees. Huseby Bruk is interested in decreasing their energy costs and increasing sustainability by creating a virtual grid. This would allow Huseby Bruk to directly use the renewable energy produced on the property. Huseby Bruk currently produces renewable energy with two sources: the historic hydroelectric plant and one set of solar PV panels on the roof of a horse stable. However, due to the current grid structure, all electricity produced must be sold to the grid instead of being directly used at Huseby Bruk.

An additional challenge is that the energy generation profile does not match demand seasonally. Huseby Bruk's energy demands are highest in the winter when more electricity is required to heat the buildings. The popular annual Christmas market at Huseby Bruk in November creates a significant additional spike in energy demand. Figure 1.3 shows electricity demand, solar production, and hydro production throughout 2024.

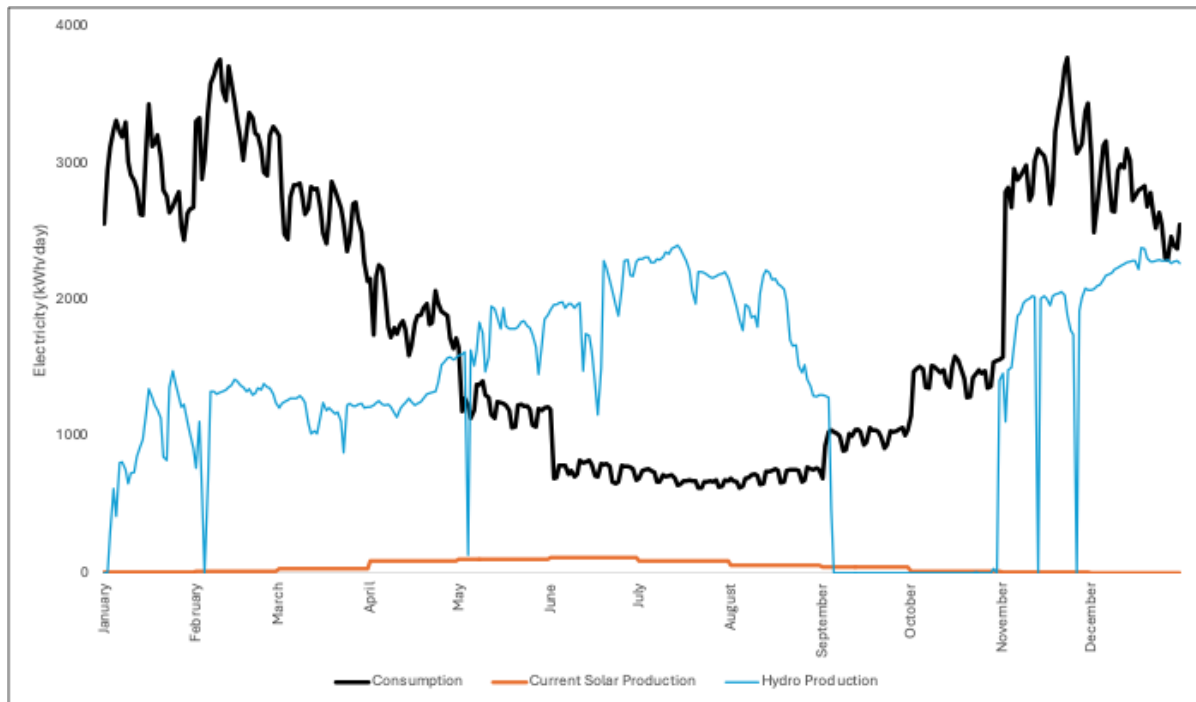


Figure 1.3. Energy production and consumption at Huseby Bruk (2024) (EO-N, 2024)

Huseby Bruk aims to be energy self-sufficient to increase resiliency, decrease costs, attract visitors, and continue Huseby Bruk's legacy of energy innovation and independence. To summarize the main goals of this project are: (1) to identify potential energy and economic savings, (2) to increase energy independence and resilience, (3) to increase the number of visitors to the property, (4) to increase sustainability, and (5) harbour and further expand on its legacy of energy innovation. In doing so, the project must be economically viable, protect and respect cultural heritage, and continue Huseby Bruk's legacy of being at the forefront of electrification.

1.2 Introduction of the Project Team and Stakeholders

The SELECT project team is a group of seven first-year EIT InnoEnergy Masters+ students. Four students are currently studying at KTH in Stockholm, Sweden and three are studying UPC in Barcelona, Spain. The names and roles of the project team members are listed in the table on page 2 of this report.

The team chose to work on this project for many reasons. The open-ended nature of this project was exciting and gave us the opportunity to shape the direction of the project. We all appreciate that the broad scope of the project allows us to research many different renewable energy topics. Furthermore, this is a real-world problem with potential for real life impact. The solutions we will propose in this project could be implemented in a tangible way. Finally, the cultural and historical nature of this project was interesting to us.

There are a variety of stakeholders involved in this project. The project is centred on Huseby Bruk, a historic property owned by Sweden's National Property Board (SFV). Sweco is a Swedish engineering consulting company that plans and designs buildings and community infrastructure. Sweco was contracted to examine the energy situation at Huseby Bruk and identify potential solutions to decrease energy costs and increase profit and sustainability. For this project we are working directly with three people at Sweco: Magnus Lindén, Emilie Andersson, and Sigrid Wikell.

The SELECT team sincerely thanks these three individuals for their support and guidance throughout the project, as well as for the insightful and enjoyable trip to Huseby Bruk.

In future work for this project, a variety of additional stakeholders would be involved. These may include Huseby Bruk's directors and employees; contractors that might do the design, construction, or maintenance of any proposed solutions; people who work with the national electrical grid; and tenants who live in the apartments at Huseby Bruk.

2 Objectives and Scope of the Project

The objective of this project is to propose a set of changes to Huseby Bruk's energy system to pursue Huseby Bruk's goal of being energy self-sufficient to increase resiliency, decrease costs, attract visitors, and continue Huseby Bruk's legacy of energy innovation and independence. This report focuses on exploring the renewable energy generation options available for Huseby Bruk, identifying potential solutions, and integrating these solutions to meet the goals of Huseby Bruk. While there is no budget for this project, the task is to show that these changes would be worth the investment and therefore be a valuable and worthwhile project for the National Property Board to pursue.

This project seeks to answer the questions:

- How can Huseby Bruk increase energy, independence and resilience?
- How can Huseby Bruk produce more renewable energy?
- How can Huseby Bruk attract tourists to the property with sustainable energy solutions?
- How can Huseby Bruk decrease energy costs?
- How can Huseby Bruk be at the forefront of energy innovation

In answering these questions, the project must be economically viable, respect the property's cultural heritage, and continue Huseby Bruk's legacy of being at the forefront of sustainable energy development.

The timeline for this project includes all major milestones and deliverables throughout the year (Figure 2.1).

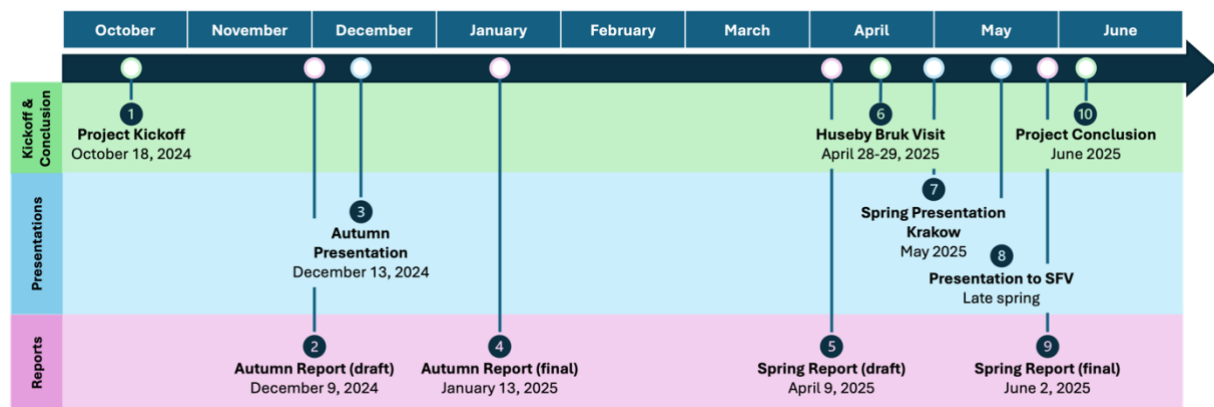


Figure 2.1. Project timeline with major milestones and deliverables

In the autumn semester (October – December) the primary objectives were to understand the problem, clarify the scope of the project, identify stakeholders, review any previous work done on the topic, and develop proposed solutions. To do so, energy consumption and production data for Huseby Bruk were analysed. Initial research was done on renewable energy generation (solar, wind, hydropower, biomass, and biogas), network structure (microgrids, virtual grids, and parallel networks), and energy storage. For each topic, considerations included feasibility, cost analysis, intermittency issues, challenges, and opportunities. The findings were reviewed with Sweco, and two initial solutions were proposed: 1) expanded solar photovoltaic (PV) production and 2) a bioenergy system.

In the spring semester (January – May), the primary objectives were to further develop the proposed solutions, consider various iterations and applications of these solutions, explore energy

storage options, and conduct an economic analysis to present a final solution. The final proposed solution includes a solar PV carport and a biogas facility. This report includes a discussion of these solutions, their advantages and disadvantages, and opportunities for future work.

3 Previous Work

Sweco has already performed some analysis regarding potential improvements to the energy system at Huseby Bruk, including the opportunity for virtual grids, parallel networks, battery energy storage, increased renewable energy production, and opportunities to further develop hydropower at Huseby Bruk. While no definite conclusions were made, the previous analysis done by Sweco provides helpful background information to guide the project. This section provides a summary of the previous work done by Sweco that is most relevant to this project.

3.1 Grid Structure

The current energy system at Huseby Bruk includes many separate grid connection points. This is expensive, as a fee is paid per connection point, and it obscures the actual degree of energy self-sufficiency of the property. Huseby Bruk currently has two sources of local energy production: solar PV panels on one rooftop and a historic hydropower plant. These sources can produce a significant amount of energy, but that energy cannot be used in other parts of Huseby Bruk due to energy sharing regulations. Electricity is sold to the grid and then Huseby Bruk must buy electricity from the grid. This is not economically or energetically efficient. Additional information about the energy network structure and energy sharing regulations is included in Appendix A.

Two solutions considered by Sweco are a parallel network, where Huseby Bruk would build its own separate electrical grid, and a virtual grid, a mathematical solution where Huseby Bruk would only have one grid connection point. A parallel network would allow Huseby Bruk to distribute the electricity produced throughout the property to meet their own demand. However, the capital cost of implementing a parallel network together with the impact it would have during construction of the property made this option infeasible. Currently, Sweco, SFV and the grid operator are looking at potential options on how to effectively solve the problem with a main consideration now that Huseby Bruk would manage its own grid and become a small-scale distribution service operator through this solution.

A virtual grid offers a promising solution to several of Huseby Bruk's current energy challenges, such as high dependence on the grid, expensive grid subscription costs, and obscured energy data. By consolidating multiple connection points into just one connection point and calculating energy flows by mathematical algorithm, a virtual grid creates a more efficient and cost-effective system. It enables prosumers (entities that both produce and consume electricity) to operate collectively as a single controlled unit (Sardi et al., 2021), thus allowing Huseby Bruk to directly use the renewable energy it generates. This integration would reduce reliance on the external grid, lower electricity costs, and provide clearer insights into energy data.

However, implementing a virtual grid is complicated. It requires advanced network design and regulatory approval, which are currently unclear in Sweden. Until recently, virtual grids were impossible in Sweden due to regulations. While recent updates to Swedish non-concessioned networks (IKN) regulations do now allow electricity sharing between buildings, this is limited to low-voltage networks and restricted within a certain spread. These changes create new opportunities for prosumers like Huseby Bruk to use or store their self-generated electricity rather than sell it to the grid but have yet to be tested. Although regulations are gradually evolving in support of such models, designing and implementing a virtual grid remains outside the technical scope of this project.

A virtual grid was determined to be a promising solution to lower grid connection costs, however, further investigation into virtual grid design is outside of the scope of this project as further work in this direction would be based more in regulations and legalities than in engineering. Instead, the focus of this project is on increasing Huseby Bruk's renewable energy production.

3.2 Battery Energy Storage and Price Arbitrage

This previous research done by Sweco also examined the potential for batteries as a tool to increase Huseby Bruk's energy self-sufficiency and as a source of income for the property through energy arbitrage services. Batteries allow any excess energy to be stored for future use, thereby decreasing the amount of electricity that needs to be purchased. Batteries can also be used to support the electricity system in the form of frequency regulation when production and consumption are not balanced. This service can be a source of income for the property. Batteries can also be used for price arbitrage; batteries are charged when electricity prices are low, and that stored energy is sold when electricity prices are high. The research conducted by Sweco found that the approximate cost of a small-scale lithium-ion battery would be around 5000 SEK per kilowatt hour (Sweco, 2024). The payback period of a battery at the time of this study was around 1-3 years, making this a promising investment. At the time of the study, the Swedish government was heavily incentivizing investments in batteries. This resulted in a lot of battery capacity coming online, which subsequently caused the market to crash. Using batteries as a source of income for Huseby Bruk is no longer viable due to the change in market conditions. Furthermore, since the mismatch in consumption and production in Huseby Bruk is mainly seasonal, batteries are not considered as an appropriate solution to make Huseby Bruk energy independent.

3.3 Hydropower

Hydropower is an essential part of Huseby Bruk's industrial history. The Helge River that flows through the property has been a critical source of power since the 1620s for iron production, milling, and electricity production. Before other power sources like the combustion engine and electricity were invented, hydropower drove industry. Thus, hydropower is important for Huseby Bruk both as an energy source and as cultural heritage.

In 2025, Sweco produced a draft study considering the current state of the hydropower plant and analysing whether it is possible to produce more electricity using hydropower at Huseby Bruk (Sweco, 2025). The system has a drop height of only three meters, but the flow rate is good, and the system can mostly be used year-round, except for when the river freezes or in long extended periods with very limited rainfall and or snowmelt. There is more than enough water flow to power an additional hydropower plant without disturbing the current production.

There are three existing water wheels at the mill and none of them are currently functional. Some of the existing water wheels could be refurbished (i.e., by replacing worn-out components such as bearings and wooden paddles) to use for milling but some are beyond restoration and efforts should be made to limit further decay. One of the water wheels could be used to power a generator if a new gear shaft was installed. Sweco estimates that this would cost 800,000 SEK and produce 5 kW or 25,000 kWh annually. Over a depreciation period of 20 years, the cost of electricity would be 1.5 SEK/kWh. This system would have a 40-year lifespan and would likely need replacement parts every 10 years. Considering maintenance costs, Sweco concludes that electricity production would almost be profitable, but the primary goal would be to "contribute to the cultural environment." To continue the analysis and obtain more reliable figures on power, electricity

production, and price, additional site visits would need to be made, measurements taken, and suppliers contacted.

In this report, Sweco also proposes optimization of the existing turbine to produce more power at lower flow rates. The existing turbine was installed in 1993 and is a Francis type turbine. Typically, these turbines can be regulated down to a flow rate of $3 \text{ m}^3/\text{s}$, however this turbine cannot be regulated to below $5 \text{ m}^3/\text{s}$. Below this flow rate, excess air is sucked in as the water level drops. If the turbine could be optimized to work at a lower flow rate, it would be able to produce power for a higher percentage of the year. However, there are only short periods when the river's flow is less than $5 \text{ m}^3/\text{s}$ so it might not make a significant difference overall.

4 Methodology

The objective of this project is to propose a set of changes to Huseby Bruk's energy system that would increase sustainability and resilience and decrease energy costs. Whether or not Huseby Bruk decides to pursue a virtual grid, the key to a more sustainable and resilient energy system is increased production of renewable energy. There are many natural resources on the Huseby Bruk property that provide potential for renewable energy such as open areas for wind and solar farms, a river for hydropower, organic waste, and feedstock for biomass or biogas.

This section presents the methodology behind the analysis of Huseby Bruk's current energy production and consumption data, and the exploration of solutions including solar PV, small-scale wind turbines, bioenergy, and energy storage.

4.1 Energy Data Analysis

In the preliminary study, the energy consumption and production data for hydro and solar power were analysed and processed to gain an understanding of Huseby Bruk's energy profile. As the obtained data from Sweco was monthly data, it had to be adjusted into hourly values to conduct a more accurate analysis.

The conversion process involved several steps. First, hourly consumption and production data for hydro and solar power in Sweden were retrieved by Sweco from Svenska Kraftnät's website (SVK, n.d.). From this data, monthly totals were calculated. These values were then used to generate weighting factors that distributed energy usage and production across individual hours. These hourly weightings were applied to Huseby Bruk's monthly data to interpolate hourly values from the available monthly values.

However, since Huseby Bruk's energy patterns differed from Sweden's national trends, additional adjustments had to be made. Weights were assigned not only by hour but also by day to better reflect local variations and ensure a smoother transition between months. Finally, to maintain consistency with the original monthly totals, a scaling factor was applied to each day if the sum of the interpolated hourly values deviated from the original monthly total.

The result of this weighing methodology is illustrated in Figure 4.1.

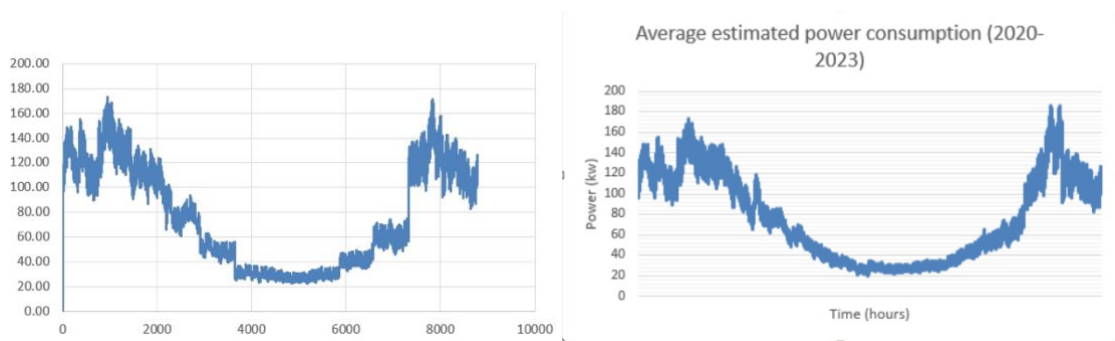


Figure 4.1. Methodology used to smoothen and correct the initially obtained hourly data (Sweco, 2025)

4.2 Electricity Power Production Data Analysis

The total hourly production (Figure 4.2) was determined by adding the hourly hydro and solar production determined using the methodology explained previously (Figure 1.3). A big drop in production can be seen in the later summer and early autumn months which can be attributed too little to no electricity production from the hydro plant during this period. This is a returning feature

which can be seen in most years, this is most likely caused by the water level of the Helge River decreasing during the summer months until a level for which the hydro plant is not operational anymore.

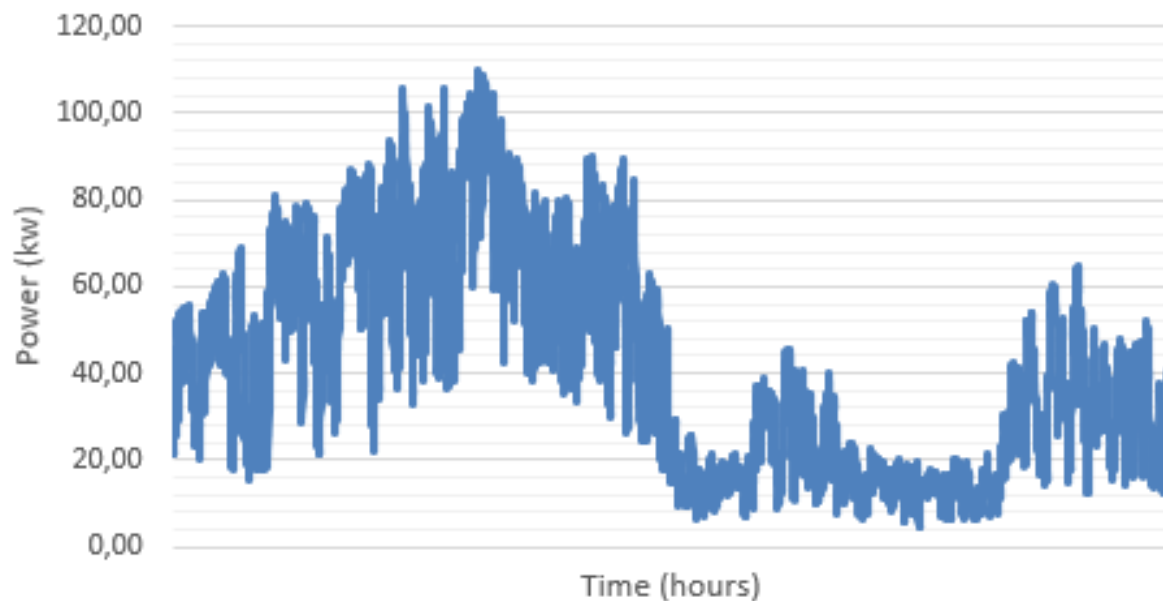


Figure 4.2. Total hourly production at Huseby Bruk (average 2020-2023) (Sneco, 2024)

4.3 Small-Scale Wind Turbines

Wind energy is one of the many renewable energy resources that Huseby Bruk has available. The installation of large-scale horizontal axis wind turbines is not feasible at Huseby Bruk due to average wind speed, high investment cost, and restrictions due to Huseby Bruk's cultural heritage.

This scenario focuses instead on the installation of micro wind, vertical-axis wind turbines (VAWT) at Huseby Bruk. Using the Global Wind Atlas, the Huseby Bruk area was determined to have an available mean wind speed of 4.49 m/s at an altitude of 10 meters. The wind speed values are within the range of 3.87 m/s to 4.75 m/s (Global Wind Atlas, 2024). The data on wind speed have been acquired at an altitude of 10 meters to satisfy Huseby Bruk's requirement of building non-invasive structures due to the property's protected status as cultural heritage.

The small-scale VAWT design was chosen considering the strict boundaries given by the property's cultural heritage, which mainly regards the habitation/market/facilities area (represented in the yellow box in Figure 4.3). It is very difficult for any proposed changes made within this area to be approved, as they change the historic atmosphere of the property.



Figure 4.3. Habitation/market/facilities area at Huseby Bruk (Google Earth, 2025)

However, during the study, a possibility of using the parking lot (outlined by the yellow box in Figure 4.4) arose and could be much more interesting, mainly because of its less strict rules.



Figure 4.4. Parking lot area at Huseby Bruk (Google Earth, 2025)

The parking lot has an area of roughly 3450 m², as measured with Google Earth's tools (Google Earth, 2025). The boundary of building non-invasive structures is here less strict, allowing to install more turbines, taller and so more performative. Small-scale VAWTs are often implemented in parking lots, sometimes together with solar PV (Figure 4.5).



Figure 4.5. WindSite's WS-030 turbines installed in a parking lot (WindSite, 2024)

However, this plan, which currently concentrates on renewable energy production, could also leverage Huseby Bruk as an educational resource and tourist attraction. The turbines could be used as a tangible model of renewable energy in action, which will thus be enhanced by the historical and cultural heritage of Huseby Bruk, which not only would be a symbol of the past, but also of the future. Guided tours for the visitors could be organized, including school groups, and interactive sessions on wind energy, its benefits, and climate change countermeasures could be arranged. Educational signs and exhibits placed in the surrounding area of the turbine would thus be able to give information about the science of wind as a power source, its role in the Swedish renewable energy goals, and the environmental effect of transitioning from fossil fuels to green energy. Working with local schools and universities would be beneficial as well, making the property a symbol of an eco-friendly environment.

Concluding, looking at the low possible energy outputs due to low wind speed, and considering the CAPEX and OPEX costs (Appendix B), installing a small-scale wind turbine system to produce electricity is believed not to be an economically viable choice. Instead, the installation of wind turbines could let Huseby Bruk attract visitors for educational purposes and could be considered in case of the property's socio-political interest. In this case, since the main purpose of such a technology (producing electricity) would not be fulfilled due to the low energy output forecasted, the investment would thus just consider the economic return given by educational trips, or seminars. In this case it could be possible to install a small number of turbines for the sake of educational opportunities and to attract visitors by showing different methods of renewable energy production and having a wide array of different renewable energy sources bundled in one place. Different models could be taken into consideration to lower the costs, mostly if the main purpose is not to generate electricity given the poor wind conditions, and so not focusing on the performances.

4.4 Solar Carport

Solar PV energy has been considered as one of the main renewable sources to generate electricity for the Huseby Bruk property because of the abundance of open farmland and available rooftops. Huseby Bruk already has one solar PV array (on the roof of the stables) that is operational and provides a small amount of energy to the property. It was a logical decision to consider the opportunity to expand solar PV production at Huseby Bruk. A solar PV system can be on-grid

(connected to grid), off-grid (with energy storage) or a hybrid system (connection to grid and energy storage during power outage).

Previously, a solar rooftop system and ground mount system using the open farmlands available to meet the Huseby Bruk consumptions were considered. Analysing the feasibility of these solutions, rooftop solar will have a lot of cultural heritage restrictions and complex interconnection issues considering massive trenching for placement of electrical cables. The ground mount system seems to be a viable option for generating a considerable amount of renewable electricity for the property, but the usage of such a big area of open land would hamper the historic atmosphere of the property and the required maintenance would be time consuming and economically stressful as well as resulting in additional security restrictions related to protecting the solar PV plant.

Additional details on these two potential system designs were included in the fall semester report submitted in December. The calculations were done in two parts: a rooftop system considering four different roofs and a ground mount system. Both the solutions were presented and the rooftop solar was not considered as a feasible solution due to the complexity of combining the solar outputs from various buildings, which would have led to complex electrical considerations and wirings along with significant amount of trenching for the laying the cables together. The ground mount system was considered to meet the yearly consumptions, but it was taking significant amount of open grassland of the property, which was hampering the property aesthetics.

Considering the importance of solar energy for Huseby Bruk but also keeping in mind the aesthetics and usage of the property, we have decided to use the available car parking lots on the property and integrate them with a solar carport. The open spaces of these parking lots are totally free from any conditions of shading which make them an ideal choice for the implementation of bi-directional solar carports (east-west orientation) (Figure 4.6). This design includes an appropriate degree of inclination for proper water drainage.



Figure 4.6. Visualization of bi-directional solar carports in Huseby Bruk parking lot

The PVWatts Calculator from the National Renewable Energy Laboratory (NREL) was used to get the actual system output per year considering annual and hourly solar radiation, DC array output, and AC energy output ([PVWatts](#)). The inputs to this calculator include the latitude and longitude of the location, the tilt of the solar carport, all kinds of system losses, orientation, DC to AC size ratio and DC capacity factor. The solution of the solar carport not only generates

renewable electricity from solar energy to power Huseby Bruk but also provides shading of the cars of the visitors. The solar carport has been designed using the whole space of one parking lot and half of the available space of the second parking lot leaving empty space for bigger vehicles. This design has been considered to maintain the uniformity of the solar carports for better inverter string arrangement and keeping the importance of aesthetics for the historic property. The final system size of the solar carport considered is shown in the Figure 4.7.

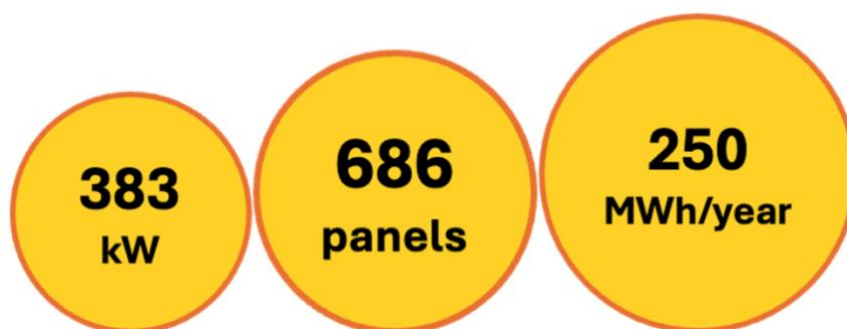


Figure 4.7. Solar carport system sizes

The economic calculations have been done considering the best-case scenario and the worst-case scenario for several system components including solar panels, inverters, mounting structures, electrical wirings, labour and installation, grid connection and permitting and other miscellaneous costs which results in a cost analysis of €421300 to €631900 as a capital investment (Table 4.1). The OPEX calculations per year have been done following the same scenarios which show approximately €8800 to €16000 in operating cost per year.

Table 4.1. CAPEX and OPEX calculation for solar carport

System Size			383 kW	
Yearly Energy Production			251884 kWh/year from PV Watts	
Daily energy Production			690.09 kWh/day	
Considering 5 hour of energy production				
Hourly Energy Production			138.02 kWh/hour	
Hourly Energy Production (each carport)			69.01 kWh/hour	
Detailed CAPEX				
Component	Cost per kW (EUR) min	Cost per kW (EUR) max	Total Cost (EUR) min	Total Cost (EUR) max
Solar Panels	250	350	95750	134050
Inverters	120	180	45960	68940
Mounting structure and carport frame	350	500	134050	191500
Electrical Wiring & BOS	80	120	30640	45960
Labor and Installation	200	300	76600	114900
Grid connection and permitting	50	100	19150	38300
Misc Costs (Design, Engineering, Prep)	50	100	19150	38300
			EUR 4,21,300.00	EUR 6,31,950.00
Detailed OPEX				
Expense Category	Cost per kW/year (EUR) min	Cost per kW/year (EUR) max	Total cost (EUR/year) min	Total cost (EUR/year) max
Maintenance and cleaning	10	20	3830	7660
Inverter replacement	8	12	3064	4596
Insurance and misc.	5	10	1915	3830
			EUR 8,809.00	EUR 16,086.00

Applying the tax incentive of Grön Teknik reduction, minimum and maximum CAPEX has been re-evaluated along with maximum and minimum OPEX, providing the net profit over a span of 25 years considering the best and the worst-case scenarios (Table 4.2). The simple payback period shows seven years for the best-case-scenario model and 10 years for the worst-case-scenario model.

The Internal Rate of Return (IRR) calculation shows about 10.83% for best-case-scenario and 3.8% for the worst-case-scenario.

Table 4.2. Total revenue, tax incentive, payback period, and IRR calculations for solar carport

Energy Production & Revenue		
Electricity Sale Price	0.2	Euro/kwh
Annual revenue (year1)	EUR 50,376.80	
Degradation Rate	0.005	
Number of years	25	years
Total Revenue	EUR 1,186,673.45	
Tax Incentive (Grön Teknik Deduction)		
Swedish Green Tech Deduction	0.194	
Min. CAPEX after Initiative	EUR 339,568.00	350,000
Max. CAPEX after Initiative	EUR 509,352.00	500,000
Payback, ROI, IRR, and NPV Calculation		
Total OPEX over 25 years (Min)	EUR 220,225.00	220,000
Total OPEX over 25 years (Max)	EUR 402,150.00	400,000
Net profit (Min Scenario) over 25 years	EUR 626,881.00	
Net profit (Max Scenario) over 25 years	EUR 275,172.00	
Simple Payback period		
Min. Case	7	years
Max. Case	10	years
IRR Calculation		
Min Case	10.83	%
Max Case	3.78	%

The designed system size with its production along with the approximate economic analysis shows the possibility and desirability of implementing solar PV carports at Huseby Bruk.

4.5 Bioenergy

Huseby Bruk is well positioned about the availability of raw materials for both biomass and biogas production. The feedstock for a biomass facility, which could include both heating and production of electricity, could be sourced from a nearby forest that is due to be cleared because of a spruce beetle infestation destroying the trees. Although the exact number of trees is not specified, the forest is extensive and contains a large quantity of usable wood. Similarly, the raw material for a biogas plant could be obtained from manure produced by the animals on the property's farm. This farm contains approximately 450 to 550 cows (Sweco, 2025). Each cow produces between 36 and 66 kilograms of manure per day (Damasceno et al., 2020), resulting in a significant and consistent supply of manure.

It was ultimately decided to move forward with the biogas plant as this solution is better suited from a logistical standpoint. As it is better to have a continuous, locally produced supply of cow manure that can be directly processed, rather than a one-time availability of biomass from a forest that must be cut down. Additionally, the wood would need pre-treatment that could not feasibly be done at Huseby Bruk, requiring a third-party stakeholder to be involved in the process. Moreover, the trees could be repurposed for other uses, whereas the manure is likely to go to waste if not converted into biogas.

Anaerobic Digester – Dimensioning

When dimensioning the anaerobic digester, two possible scenarios are considered: exploit the raw material (cow manure), thus maximizing the biogas production, or adapt its size to match the property's energy demand. The first scenario opens the possibility of selling the surplus production; the second, however, requires a storage system.

If maximizing the raw material's potential, the following calculations result:

Feedstock (kg/day)

$$\text{Feedstock (worst case scenario)} = 450 \text{ cows} * 36 \frac{\text{kg}}{\text{day}} = 16200 \frac{\text{kg}}{\text{day}} \quad (1)$$

$$\text{Feedstock (best case scenario)} = 550 \text{ cows} * 66 \frac{\text{kg}}{\text{day}} = 36300 \frac{\text{kg}}{\text{day}} \quad (2)$$

Feedstock [m³/day]

The manure density was determined to be in a range between 550 to 800 kg/m³ (Husted, 1996).

$$\text{Feedstock (worst case scenario)} = \frac{16\,200 \text{ kg/day}}{800 \text{ kg/m}^3} = 20.25 \frac{\text{m}^3}{\text{day}} \quad (3)$$

$$\text{Feedstock (best case scenario)} = \frac{36\,300 \text{ kg/day}}{550 \text{ kg/m}^3} = 66 \frac{\text{m}^3}{\text{day}} \quad (4)$$

Hydraulic retention time (HRT)

$$\text{HRT} = 30 \text{ days}$$

The organic material remains in the reactor for a specific period during which anaerobic digestion occurs. The average duration the material stays in the reactor before being discharged is known as the hydraulic retention time (HRT). For a case in Sweden, HRT normally varies between ten and forty days (Nordberg et al, 2005). An HRT of 30 days has been retained enough for this application.

With the cited data, it is possible to make a rough estimation of the digester's size needed (Johnston et al, 2020):

Digester volume V_{DG} [m³]

$$V_{DG}(\text{worst case scenario}) = \text{Feedstock (worst case scenario)} * \text{HRT} = 20.25 \frac{\text{m}^3}{\text{day}} * 30 \text{ days} = 607.5 \text{ m}^3 \quad (5)$$

$$V_{DG}(\text{best case scenario}) = \text{Feedstock (best case scenario)} * \text{HRT} = 66 \frac{\text{m}^3}{\text{day}} * 30 \text{ days} = 1980 \text{ m}^3 \quad (6)$$

These calculations are simplified, and do not consider various steps in the biogas-making process, especially the pre-treatment of the manure. However, these data can be taken as an estimation and a reference, to in case further implement and design into the details the system.

As output, the gas produced by the manure, not purified, can be estimated as:

Gas production potential

The content of raw biogas per kg of manure varies between 0.023 and 0.040 m³/kg (Upadhyay, 1984).

$$\text{Gas production (worst case scenario)} = 16200 \frac{\text{kg}}{\text{day}} * 0.023 \frac{\text{m}^3}{\text{kg}} = 372.6 \frac{\text{m}^3}{\text{day}} \quad (7)$$

$$\text{Gas production (best case scenario)} = 36300 \frac{\text{kg}}{\text{day}} * 0.040 \frac{\text{m}^3}{\text{kg}} = 1452 \frac{\text{m}^3}{\text{day}} \quad (8)$$

After the gas is produced, purification is crucial to increase the energy content per unit volume and to eliminate the corrosive effects of hydrogen sulphide (H₂S). Purification involves the removal of carbon dioxide (CO₂), H₂S, and water vapor, enabling the biogas to be compressed into cylinders for storage or transport. By reducing the concentration of CO₂, the methane content of the gas is increased, thereby raising its heating value. Additionally, removing CO₂ not only enhances fuel quality but also reduces greenhouse gas emissions, in case they are not directly released to the atmosphere (deTar, 2013). Raw biogas produced from digestion is roughly 60% methane and 29% CO₂ with trace elements of H₂S and 3,1% of water vapor. Finally, the purification of raw biogas removes roughly 55 to 30% of its volume (Wikipedia, 2024).

$$\text{Biomethane (worst case scenario)} = (1 - 0,55) * 372.6 \frac{\text{m}^3}{\text{day}} = 167.67 \frac{\text{m}^3}{\text{day}} \quad (9)$$

$$\text{Biomethane (best case scenario)} = (1 - 0,3) * 1452 \frac{\text{m}^3}{\text{day}} = 1016.4 \frac{\text{m}^3}{\text{day}} \quad (10)$$

The last main step is the storage of biomethane. Storing biogas presents challenges typical of gaseous fuels. Unlike some gases, biogas cannot be easily liquefied at ambient temperatures, as its critical temperature and pressure are –82.5°C and 47.5 bar, respectively. Therefore, it remains in gaseous form under normal conditions. Compressing biogas helps reduce storage volume, concentrate its energy content, and provide the pressure needed to overcome flow resistance. Common storage solutions include repurposed propane or butane tanks and commercial gas cylinders, typically rated for pressures up to 200 bar (deTar, 2013) (Figure 4.8).

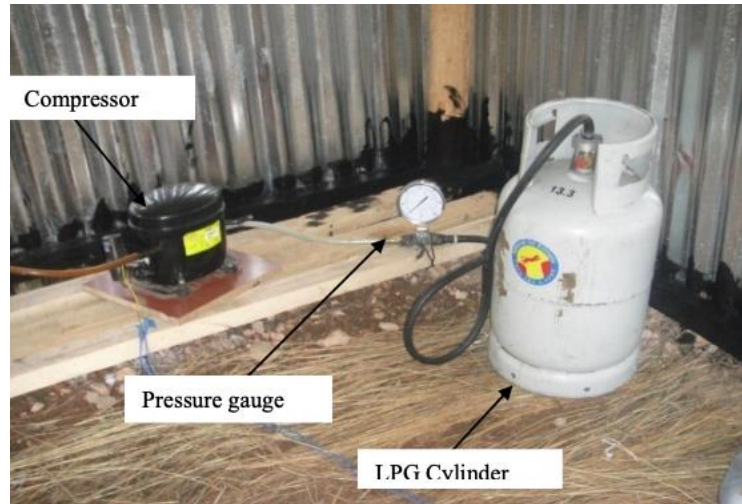


Figure 4.8. Biogas compression and storing (J. E. deTar, 2013)

Anaerobic Digester – Costs

The cost of constructing an anaerobic digester depends on various aspects. The CAPEX depends on the size, number of livestock, redundancy and phasing of the plant, technology used, automation of the process and aimed product's purity. Moreover, annual operation and maintenance costs, such as reparation, cleaning, labour, transportation of feedstock, consumables (polymers, filter medias, etc.), laboratories analysis and insurance plus taxes, must be taken into consideration (Marte et al, 2009). The OPEX costs are estimated between 1 and 10% of the CAPEX (BiogasWorld, 2016).

All the costs should also relate to the technical lifetime of a biogas plant, estimated between 15 and 25 years (BiogasWorld, 2016). To develop an economically feasible biogas production from manure, a key point is to improve the degradation and methane yield, as well as reducing risks for pathogen spreading (Ahlberg-Eliasson et al, 2021).

The CAPEX for an anaerobic digestion facility, ranging from 50 to 400 kW installed combined heat and power (CHP) electric power, can be calculated as (Koutinas et al, 2024):

$$CAPEX \text{ (low cost scenario)} = V_{DG \text{ worst case}} * [1000 - 40 * \ln V_{DG \text{ worst case}}] = 515,529 \text{ €}$$

$$CAPEX \text{ (high cost scenario)} = V_{DG \text{ best case}} * [1000 - 40 * \ln V_{DG \text{ best case}}] = 1,378,804 \text{ €}$$

It includes the engineering, procurement and construction of various components, such as a buffer tank with a mixer and pumps, a digester tank with a mixer, heat exchanger and a membrane gas holder. Additionally, biogas equipment comprising a scrubber, dryer, and CHP unit was accounted for, along with monitoring equipment including liquid and gas flow meters, temperature, pressure and level sensors. Safety equipment such as flare, pressure relief valves, flame arresters, and explosivity sensors were also included (Koutinas et al, 2024).

Subsidies for Biogas

To promote the installation of a biogas digester and a biogas heat generator, subsidies from the Swedish government or the European Union (EU) must be taken into consideration.

A regulation currently exists in Sweden providing aid for non-food biogas and bio-propane used for heating or as motor fuel. The plan is in concordance with the REPowerEU plan (European Commission, 2024), established after the Russia's invasion of Ukraine in 2022, which aims to

increase the energy independence of the EU and limited dependence on gas from non-EU countries.

The Swedish plan is approved by the EU Commission until 31 December 2030, and it includes two schemes that exempt taxation from energy and CO₂ biogas and bio-propane that is used in heat generation, and biogas and bio-propane that is used as motor fuel. Corresponding tax 2023 on natural gas is 3946 SEK/1000 Nm³ (~33 €/MWh). Moreover, maximum 0.40 SEK/kWh (~35 €/MWh) is given as support for biogas produced from manure (Energigas Sverige, 2024); maximum 0.30 SEK/kWh (€0.02/kWh) if the biogas is upgraded to the same quality as natural gas; liquefied biogas can receive an additional 0.15 SEK/kWh (€0.01/kWh) in support (Bioenergy News, 2024).

The aims of the regulation are the following (Council of the European Union, 2024):

- “Under the heating fuel scheme, the direct beneficiaries of the tax exemptions are the end users (e.g., companies active in the heating, district heating or manufacturing sectors). In this case, the end user may deduct the tax in its own tax declaration (if it is a taxpayer of the energy and CO₂ taxes), but it is also common that the end user acquires the gas inclusive of tax and applies for a refund.
- Both schemes aim at reducing the final price of biogas and bio-propane for end-consumers. The schemes also lower the costs of consumption for producers that produce biogas (and bio-propane) for their own use.
- While the direct beneficiaries of the schemes are the undertakings that claim the tax exemptions to the tax agency (either directly for the taxpayers or as a refund for the end users in the heating fuel scheme), the schemes indirectly benefit producers of sustainable biogas and bio-propane. They indirectly benefit from the schemes through an increased demand for their products (sustainable biogas or bio-propane) from final consumers.”

The expenses related to the installation of the biogas digester, and biogas heat generator are covered by another Swedish investment programme. This investment supports, covering approximately 45–65% of eligible costs, all types of projects or measures that result in significant reductions in greenhouse gas (GHG) emissions during the period 2015–2028. For 2024, the allocated budget is 4.1 billion SEK per year (approximately €0.36 billion). A substantial share of this support has been directed toward biomethane-related investments, including numerous biogas plant developments (Energigas Sverige, 2024). Official information regarding the application to the incentives could be made through the Nature Conservation Agency website. (Naturvårdsverket, 2025)

Finally, if building a digester to produce biogas, the incentives would be addressed to the kWh of energy produced; if building a biogas heat generator, the regulation is helping with the energy and taxation mitigation of CO₂, which is set to zero.

4.6 Battery Energy Storage

To address the inherent intermittency of renewable energy generation and increase energy resiliency, the feasibility of implementing an energy storage system (ESS) at Huseby Bruk was evaluated. A model based on the methodology proposed by Ren et al. was used to optimally size the ESS across multiple time scales—yearly, monthly, weekly, and daily—with the objective of maximizing renewable energy utilization and minimizing storage capacity. The approach relies on analysing cumulative charge and discharge patterns in response to variations in generation and load

demand; storage capacity is then sized according to the largest cumulative charge or discharge observed (Ren et al., 2023). This section focuses only on the intraday (daily) design as it is the most practical and cost-effective option. Sizing results for other timescales are available in Appendix D.

The analysis includes solar generation from the proposed solar carport, existing renewable sources, and hourly consumption data. The resulting daily storage profile is shown in Figure 4.9. During winter months, no storage is required since demand is high and solar generation is low, meaning all generated energy is consumed immediately. The amount of energy stored begins to rise in March when solar output increases and demand falls, reaching a peak capacity of 390 kWh on March 18. In summer, frequent surplus generation leads the ESS to fill to its maximum capacity, discharging primarily at night when solar production ceases. Over the duration of a year, the total energy discharged from the optimally sized system amounts to 13 MWh.

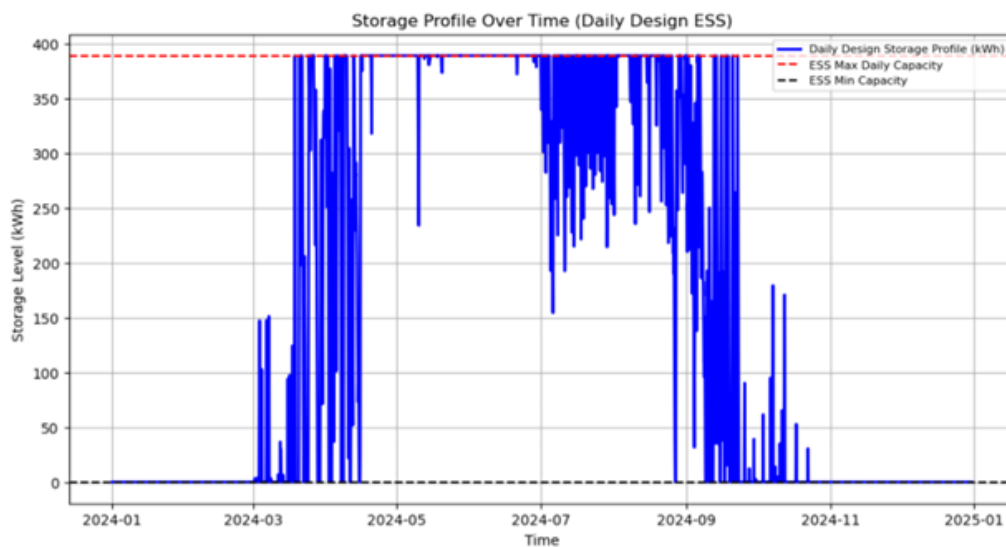


Figure 4.9. Storage profile using daily design period

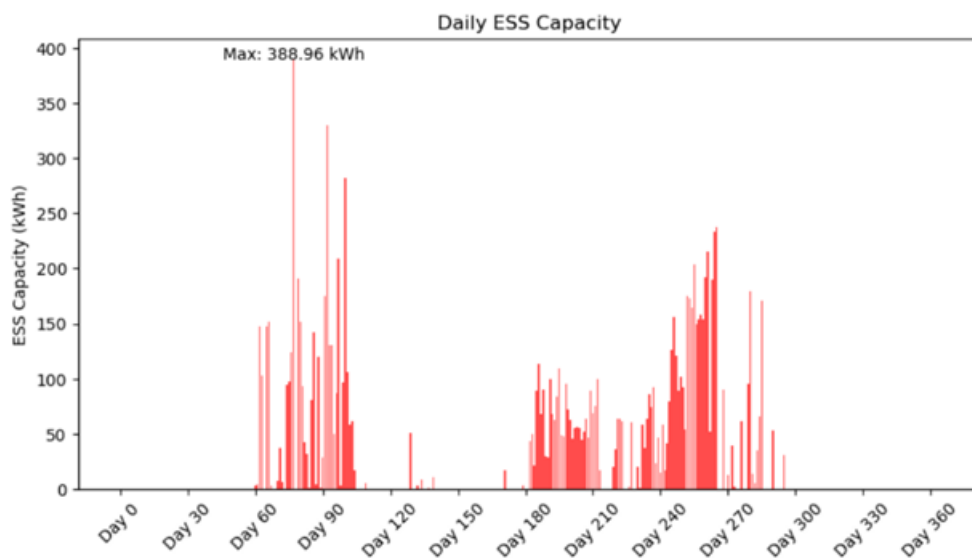


Figure 4.10. EES storage size calculated using daily design period

Following capacity determination, a cost analysis was conducted assuming a 4-hour commercial battery system with a 15-year lifespan. Based on 2025 data from NREL, the CAPEX is estimated

at 20 SEK/W, with a fixed OPEX of 475 SEK/kW-year (47 USD/kW-year). This results in a total investment of approximately 2.09 million SEK. Another crucial metric, levelized cost of storage (LCOS), is determined considering total investment cost for a system and the total amount of electric energy that this system delivers during its lifetime, equal to 11 SEK/kWh. This outcome indicates a relatively high cost compared to the electricity price in Sweden, which stands at about 2 SEK/kWh (GlobalPetrolPrices.com, 2024), while the battery storage can only provide approximately 2% of the annual electricity demand. Moreover, according to the battery market research from Sweco, the market of auxiliary service crashed last year, making it difficult to gain revenue from battery storage. In conclusion, battery storage is not an economically sound option for Huseby Bruk.

5 Presentation of the Project Solution

Two renewable energy generation solutions for Huseby Bruk's future energy system are presented here: 1) a solar PV carport and 2) biogas production. These solutions are independent, and each could be implemented separately. However, the integration of the solutions would be ideal for Huseby Bruk.

5.1 Solar Carport

The first proposed solution for additional renewable energy generation capacity at Huseby Bruk is a solar carport. Solar carports are becoming popular in Sweden as an innovative, sustainable solution that combines parking lots with renewable energy generation. These structures not only provide covered parking but also generate solar energy from the PV panels attached to it. The solar energy can be converted to electrical energy using inverters to power adjacent buildings or charge electric vehicles (EVs) (Kuchta, 2022).

Huseby Bruk has two parking areas, each having an area of about 3500 m². The solar carports are currently designed using the whole space of one parking lot, while the other parking lot only half of the area is utilized for carports leaving the remaining space for larger vehicles like trucks and caravans that would not fit underneath the solar carport (Figure 5.1).



Figure 5.1. Using Google Earth Pro to show the parking lots available for solar carports

The solar carports have been designed as Y-shaped dual sloped E-W facing (260 degrees west and 80 degrees east) (Figure 5.2). Where one parking area is fully filled with the carports so PV generation and car shading and the other parking area is left bare partly to accommodate for bigger vehicles.



Figure 5.2. Y-shaped dual sloped E-W facing (260 degrees west and 80 degrees East) (PV Europe)

Both the designs have been done considering an area of about 5100 m² with 6 degrees of slope. Both the carport designs consist of 696 modules with a system size of approximately 383 kW, with yearly calculation coming to 252 MWh/year which is approximately 74% of the assumed yearly required consumption of Huseby Bruk (342 MWh/year). Figure 5.3 and Figure 5.4 show the dimensions of the system design using AutoCAD. Results of yearly power output calculations were done using the NREL PVWatts Calculator and further discussion of these results are included in the appendix sections of this report.

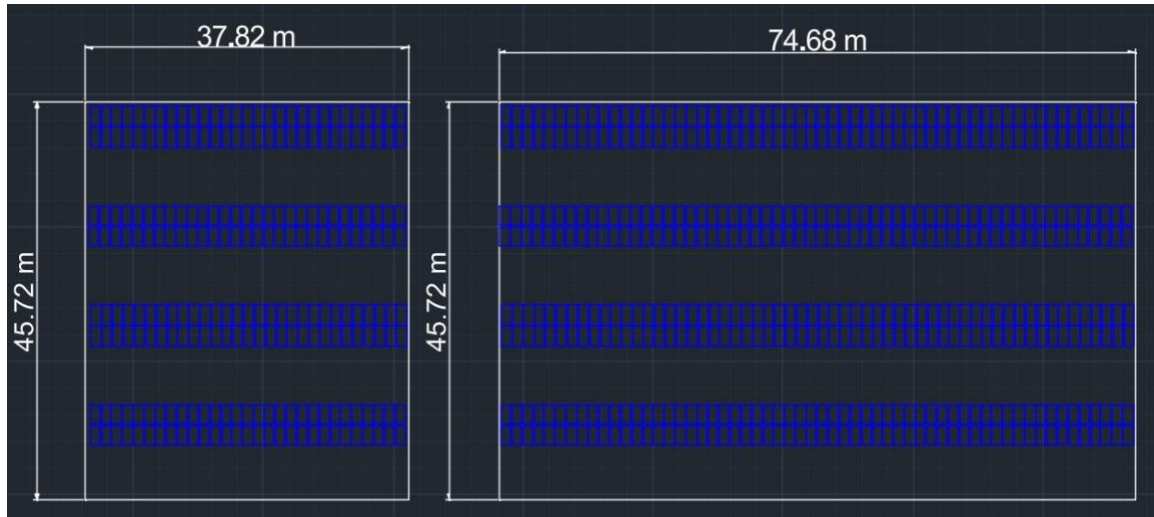


Figure 5.3. Single-sloped West facing solar carports with 696 modules

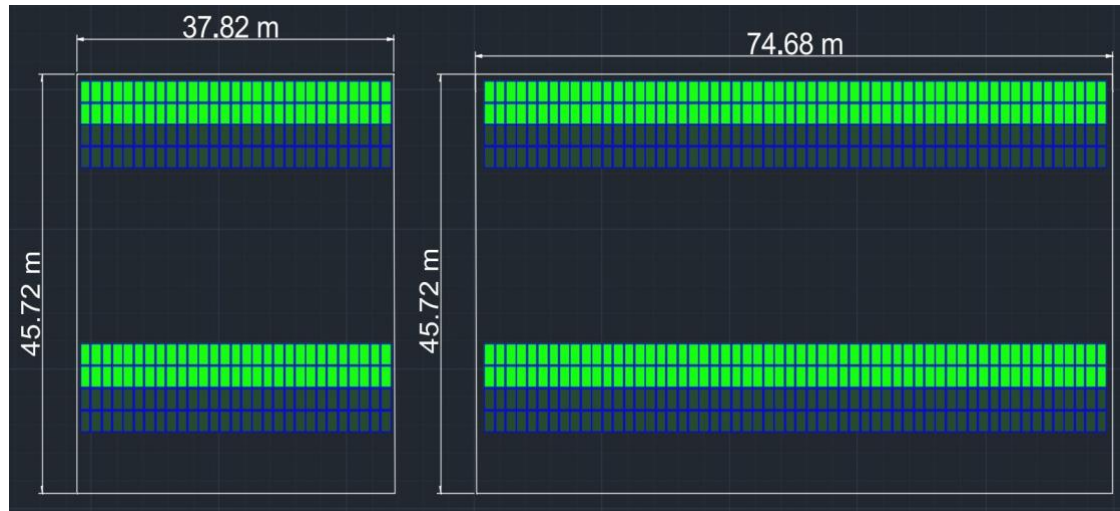


Figure 5.4. Y-shaped dual-sloped) solar carports with 696 modules (PV Europe)

The solar carports designed have shown a significant amount of energy production during the summertime, exceeding the demand expectation of Huseby Bruk but during the winter days the output of the solar PV carport has been below the demand load of Huseby Bruk. Figure 5.5 shows the characteristic of the PV graph over the whole year.

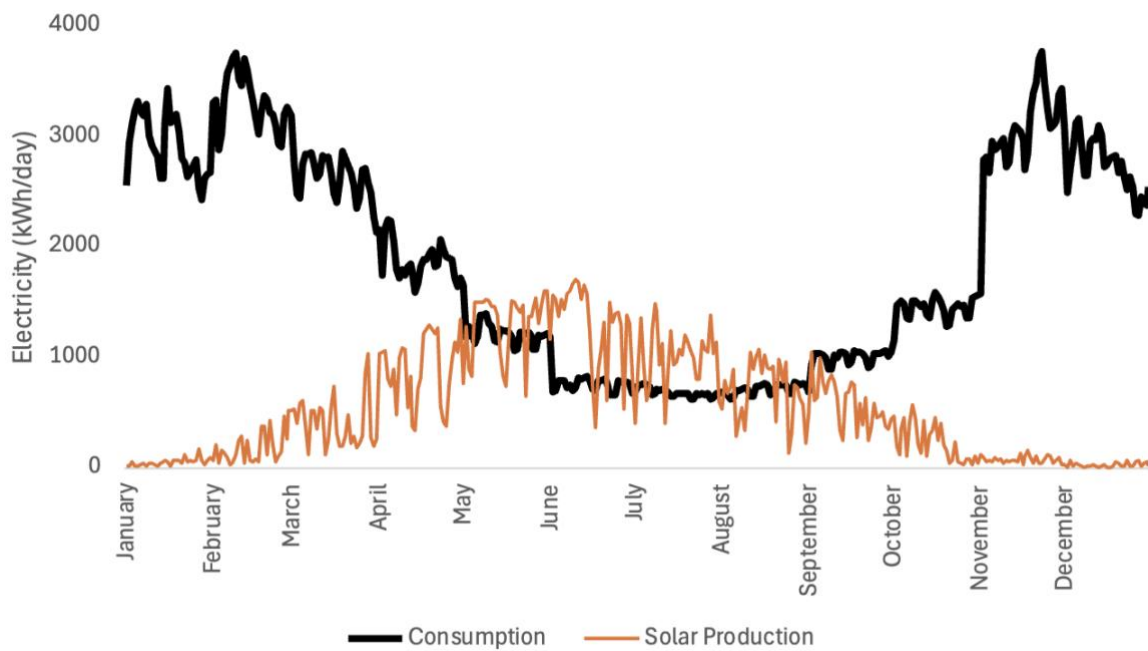


Figure 5.5. Yearly PV production vs. hourly Huseby Bruk consumption

5.2 Bioenergy

Given the rural location of Huseby Bruk, biogas places itself as an interesting opportunity for Huseby Bruk. Indeed, it can use locally sourced cow manure, thus enhance circular economy and strengthen Huseby Bruk's energy independence.

Anaerobic Digestion – Process Stages

Developing a biogas digester using cattle manure involves several stages. Firstly, the manure needs to be collected and transported where the biogas facilities are placed. During this first stage, and after being collected, the solids need to be separated from the liquids and grinded, thus enhancing the digestion efficiency. This process is typically achieved with mechanical separators, like screw presses and centrifuges. Secondly, the raw material is introduced into the digester, where the core process happens. The digester must be maintained at fixed condition (temperature, pressure, free oxygen), thus requiring energy.

Raw biogas contains impurities like CO₂, H₂S, and moisture, which require removal to produce biomethane. Each purification treatment requires energy, water and/or chemicals. Once purified, the biomethane can be utilized, after having it transmitted via pipelines or stored into gas bottles.

Beyond the cited main steps, further processes should be considered. In fact, the residual material post-digestion, which is rich in nutrients can be again utilized as a biofertilizer. Moreover, implementing automated systems to monitor temperature, pH and gas composition could ensure optimal operation of the plant, maximizing also the investment.

Biomethane – Applications in the Properties

The stored biomethane can be used for different applications. For Huseby Bruk's case, it could power a bio-fuelled heating system, generate electricity or sell it into the Swedish market.

The Swedish biomethane market is largely off-grid, characterized by numerous small-scale local and regional distribution networks, as well as stand-alone biogas plants and refuelling stations. A significant portion of biomethane in Sweden is transported by road in compressed form, typically at 200 to 260 bars. The gas pipeline infrastructure is relatively limited, concentrated mainly in the south-western region of the country, where the transmission network connects to the European gas grid via the Dragör exit point in Denmark. Additionally, a separate regional gas network operates in the Stockholm area. Right now, no system for guarantees of origin (GO) for biogas is currently working (Energigas Sverige, 2024). However, given the property's limited experience and technical knowledge in this sector, the option of selling the produced biomethane on the market has been excluded.

The first considered possibility involves using a boiler fuelled by biomethane and a respective water-based heating system (also known as a hydronic heating system). Many modern gas boilers are compatible with biomethane due to its similar properties to natural gas, so identifying a suitable option on the market would be relatively feasible.

Switching from the current electric resistance heating system at Huseby Bruk to hydronic heating is technically a viable option, as water systems provide improved comfort and more efficient heat distribution compared to simple metal heating element. On the other hand, a complete system replacement would result in a high investment for the property with no return on investment, as well as long timelines for reconstructions and obtaining permits from residents. Furthermore, the section from boiler to end user in a water-based scenario presents more losses (both thermal and energetic, since it needs a water pump) than the electricity solution. The heating losses in the pipes,

which can be mitigated with proper coatings, can account for a significant percentage of the total ones, as shown in Figure 5.6. Finally, a water-based system would need to produce more energy (kWh(th)) than with the actual electricity system (kWh(e)) at a cost of more discomfort inside the heated areas. Nevertheless, a case study was made to assess the feasibility of these considerations.

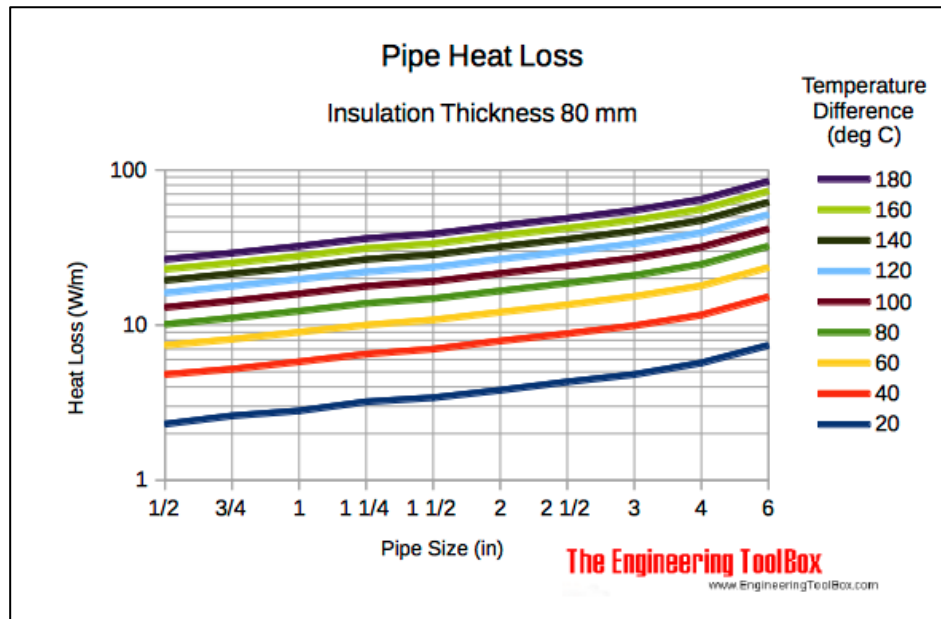


Figure 5.6. Heat loss in insulated pipes by size and temperature differences The Engineering ToolBox, n.d.)

As stated previously, the heating demand in Huseby Bruk represents the main challenge to assess energy optimization (producing in sync with demand). The proposed solar PV system, being able to generate abundant energy during the summer months, produces less in winter, when heating demand is at its peak. This discrepancy creates a need to find alternative heating solutions to maintain a stable indoor temperature. To address this, various storage solutions were studied to stock the surplus energy in warm months until colder periods. As explained in Section 4.6, no feasible options were identified that could store energy for such a long time.

The manor house (*“slottet”*) was initially targeted to modify the current heating system, since it was identified as the most energy-consuming building. The idea was to replace its heating system while leaving the other buildings as they were. The approach was considered quantitatively reasonable, as it would provide most of the property’s energy without changing the system in all the other heating infrastructures.

However, after receiving more information from Sweco on the actual heating system, the proposal was later rejected. The primary reason for this decision was the unique heating infrastructure in the manor house. In fact, the building utilizes traditional tiled stoves, containing built-in electric resistance heaters. Due to the historical heritage and significant attractions and museum in the manor house, modifying the system into a water-based one was considered impractical, as preserving the building’s integrity and original heating infrastructure was a priority.

This outcome reinforces the need to explore building-specific strategies that respect both historical constraints and energy optimization goals by generating electricity instead of heating.

Electricity Generation from Pure Biomethane

Once biogas is upgraded to biomethane, it becomes a high-quality fuel that is essentially equivalent to natural gas in terms of how it behaves. That makes it suitable for producing electricity using

systems like internal combustion engines (ICEs), gas turbines, microturbines, or even fuel cells, without needing major modifications (Kumar et al., 2022). These technologies, which are common on the market, are modular and can be scaled depending on the size of the installation, so they can work for both small properties and bigger setups.

According to the International Energy Agency (IEA), almost two-thirds of biogas production in 2018 was used to generate electricity and heat, with an approximately equal split between electricity-only facilities and cogeneration facilities (IEA, n.d.). Around 30% was consumed in buildings, mainly in the residential sector for cooking and heating, with the remainder upgraded to biomethane and blended into the gas networks or used as transport fuel.

Unlike CHP systems, which can become complex and expensive, especially when the heating setup is already electric, using biomethane just for electricity is a simpler and more cost-effective option. In fact, there are real-life examples where using biomethane in full-electric setups led to decent CO₂ savings compared to just relying on the national grid, especially in countries with dirtier energy mixes (Negro et al., 2024). Furthermore, by upgrading to biomethane, there is the possibility of getting cleaner fuel that works better in more advanced tech like fuel cells, while it is easier to store and transport when compressed (Kumar et al., 2022).

Looking at the numbers, based on the estimated biomethane production potential (between 167.7 and 1016.40 m³/day), and scaling that up to yearly by simply multiplying by 365 days, the site could be classified as a small-medium biomethane producer. That rules out more complex and expensive technologies like microturbines or gas turbines. A simple gas-powered ICE, on the other hand, fits perfectly. Assuming an average efficiency of 35% (Wikipedia, n.d.) and using the LHV of biomethane as 36 MJ/m³, or around 10 kWh/m³ (IEA, n.d.), daily electricity production could range from 24.45 kWh to 148.23 kWh depending on gas output and quality. Over the year, that's between 214.2 MWh(e) and 1298.5 MWh(e) of clean, local power.

What really makes this setup interesting is the flexibility and ability to meet seasonal demand. Since energy demand spikes in winter mainly because of heating, being able to switch on the ICE during peak periods is a huge plus. It could act as a backup or a peak load solution, especially when the solar panels are not delivering enough energy, making it a flexible option. That way, the property is more energy self-sufficient and the need to purchase electricity from the grid is reduced. Furthermore, the waste products of the anaerobic digester can be used as a fertilizer on different parts of the farm.

5.3 Integration of Solutions

Strategic Benefits of a Hybrid Energy System

The combination of solar carports and bioenergy-generated electricity creates an adaptive and resilient energy system crafted to suit Huseby Bruk's unique position. As already explained, the two systems can operate separately; however, the combination enables the property to assist with the seasonal and hourly mismatches between energy demand and production, generating a robust system. Solar energy production peaks in the summer months, when the property's demand is moderate. During those long, dark Swedish winters, when heating and lighting are in high demand, solar output drops. The mismatch makes biogas a suitable component because energy production from cow manure can run all year round and be stored when needed. The flexibility of storing and dispatching energy when needed can maintain stable operation even in the harshest weather.

Moreover, this integrated system offers advantages in many other aspects besides technical performance. Geopolitical uncertainties, with the ongoing war in Ukraine and the growing tensions with Russia, create a landscape in which the dependability on local renewable resources increases energy security for Huseby Bruk. The Swedish grid is generally stable and clean; however, a local generation could reduce any threats to blackouts or market price fluctuations.

Scenario Analysis: Adaptive vs Constant Biogas Generation

To decide on the viability of adding biogas to the system, the production range of its potential was considered. As already mentioned, biomethane production at Huseby Bruk is estimated to be between 210 MWh(e) and 1298 MWh(e) per year, depending on various factors such as availability of manure, digestibility efficiency, and gas quality. Taking the current production profile into account, the missing demand between production and consumption is 260 MWh/year. This being a conservative guess, the bioelectricity production was modeled to match the missing demand as shown in Figure 5.7.

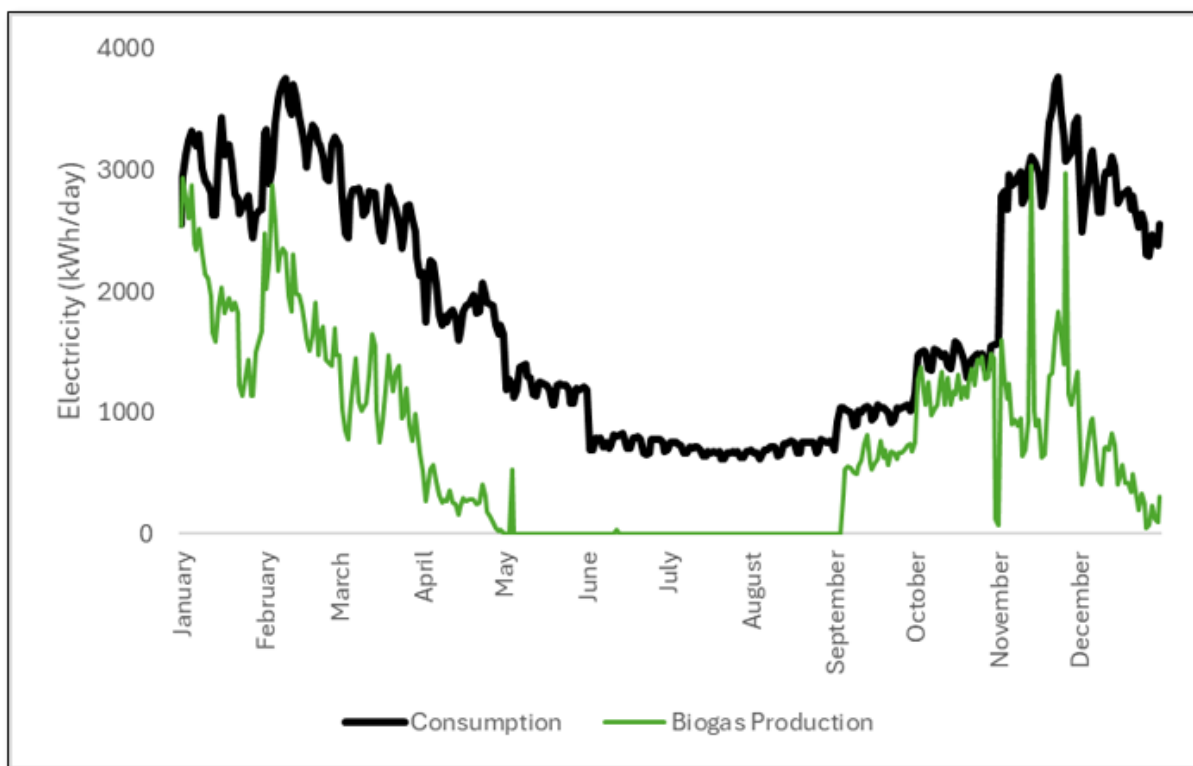


Figure 5.7. Bioelectricity modeled to meet demand (incl. solar carport production)

Two integration scenarios were modelled to understand the system's performance under different biogas generation strategies. In the first scenario, biogas electricity generation is used in a flexible way, activated only when solar and hydro are not sufficient to meet the demand. This "adaptive configuration" allows the property to reach zero grid dependency, covering the entire annual demand with local renewable sources. However, to ensure that even peak hourly loads can be met, the internal combustion engine (ICE) used to convert biomethane to electricity must have a capacity of 155 kW. A modular system composed of several medium-scale engines could be ideal, as it offers greater generation flexibility and ensures continuous electricity production even during maintenance of individual components. This ensures coverage even during high-demand hours, though it would not be fully utilized throughout the year.

In the second scenario, the biogas generator would be running continuously at full 29.6 kW output. This way, electricity could be generated steadily, and operations would be simpler, at the cost of not covering all demand peaks, leaving a residual energy requirement of about 127 MWh/year. Nevertheless, the system still produces a large annual surplus of 420 MWh, meaning that local generation exceeds total demand by a wide margin.

These two scenarios are summarized in Table 5.1, which compares the total annual surplus and grid energy need in both configurations, with and without biogas. The “adaptive” scenario (matching production with demand) eliminates grid reliance but requires a higher-capacity generator. The “constant” scenario simplifies operation and still drastically reduces dependence on the grid.

*Table 5.1. Comparison of energy surplus and grid need in different scenarios
(Source: Internal calculations based on solar, hydro, and biogas production models)*

	Yearly Surplus (in MWh)	Yearly Energy Need (in MWh)
Baseline Scenario: PV + hydro	294.7	262.2
Adaptive Biogas + PV + hydro	294.7	0.0
Constant Biogas + PV + hydro	697.3	21.8

Additional Opportunities and Long-Term Outlook

In addition to the solar carports and biogas, Huseby Bruk can also consider the installation of small-scale wind turbines. VAWTs offer a small design that can be installed near the carports or in parts of the estate without ruining the surrounding landscape or compromising its cultural attributes. While they would not generate much energy, they would be capable of generating utilisable electricity on windy days when sunlight performance is compromised (or non-existent) and help balance the system under most weather conditions. These installations, not insignificantly, could also serve an educational purpose. As a cultural and heritage site attracting large numbers of visitors annually, Huseby Bruk can possibly bring forth renewable energy alternatives without compromising its character. The small wind turbines can demonstrate how different technologies fit into, and are part of, a more sustainable future, offering educational visits for schools, tourists and interested visitors.

Overall, the integration of solar PV, bioenergy, and small-scale wind power, alongside the already existing hydroelectric plant, provides a strong base for an energy system that is resilient, modular and local. The technical configuration supports significantly increased energy autonomy, reduced carbon footprint and alignment with Swedish and European sustainability goals. The proposed solution offers a scalable and adaptable model that respects the property’s cultural heritage while embracing, once again, energy innovation.

6 Discussion

The goal of this project is to attract visitors to Huseby Bruk, reduce energy costs, and increase sustainability by creating a more efficient, resilient clean energy system and continue Huseby Bruk's legacy of energy innovation and independence. The proposed solution – a solar PV carport, small-scale VAWTs and biogas production – achieve this goal by producing clean energy using Huseby Bruk's natural resources. The implementation of all the above solutions would bring the greatest impact to Huseby Bruk's energy system.

6.1 Solar Carport

The proposed solar solution is to expand Huseby Bruk's solar PV generation capacity through the addition of solar carports. The property already has one functional solar PV system and there is potential to introduce more solar power to increase renewable energy generation.

Huseby Bruk's location in the southern part of Sweden means that it gets more sunlight throughout the year than in other parts of the country. The location of Huseby Bruk has around 1000 kWh/m² global radiation (Figure 6.1). The widespread area of the car parking inside the property makes it an ideal location for a solar PV carport system.

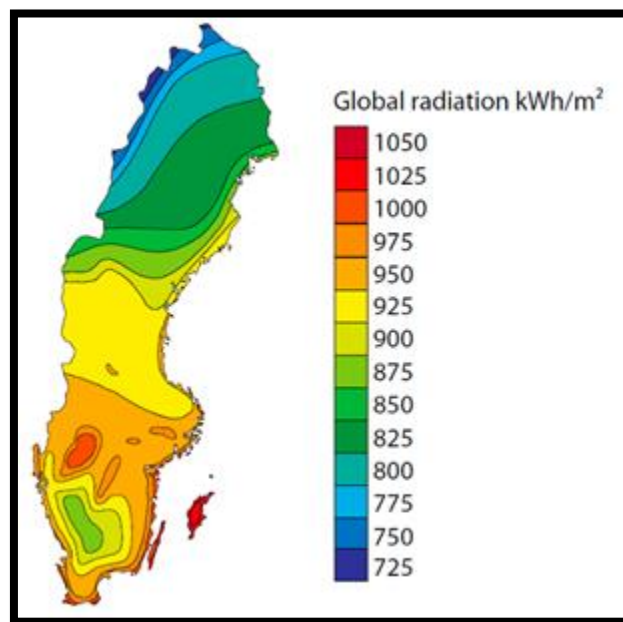


Figure 6.1. Map of Sweden showing global radiation (Westerberg, 2023)

The idea of this solution is to put more solar panels in the available car parking areas at Huseby Bruk. Considering the open space available without any kind of shading, solar panels will have a moderately good yield, having an average of 3 PSH facing west and 2,6 PSH facing east-west (Turky, 2024). Table 6.1 shows the PSH and optimal tilt angle for Växjö, an area close to Huseby Bruk.

Table 6.1. Växjö's Peak Sun Hours over the seasons along with the best possible tilt angle (Westerberg, 2023)

Location	Subdivision	Latitude	Longitude	kWh/day Summer	kWh/day Autumn	kWh/day Winter	kWh/day Spring	Panel Tilt Angle
Vaxjo	Kronoberg County	56.8746	14.8124	5.68	1.77	0.80	4.28	47° South

The proposed solar carport design sums up to approximately 383 kW. The total production coming from the carport solar will add up to 252 MWh/year which is around 75% of Huseby Bruk's yearly electricity consumption.

The importance of the solar carports will not only help Huseby Bruk to get electricity from the renewable source of solar energy but will also help the visitors to park their vehicles under the shade and appreciate the effort of Huseby Bruk to generate energy from sustainable energy.

Swedish Solar Market

Solar energy in Sweden is experiencing an upward trend. Looking at data of installed grid connected solar PV in Växjö (Table 6.2), 2464 solar PV systems were installed in 2023, a significant increase from the 1402 installations installed in 2022.

Table 6.2. Grid-connected solar PV systems in Växjö (energimyndigheten, n.d.)

	2022	2023
< 20 kW	1278	2260
20 kW – 1 000 kW	124	204
> 1 000 kW	0	0
Total	1402	2464

The Swedish solar market has also shown around 75% increase in commercial solar installations (PV Europe, 2024). An estimated size of 460 MW solar has been installed in Sweden during the first half of 2024 (PV Europe, 2024). To support the PV business model, some subsidies and a feed-in premium scheme have been exclusively allocated to distributed PV installations. There has been a constant upward trend of data for PV installations (off-grid, on-grid distributed, and on-grid centralized) since the subsidies have been introduced (Westerberg, 2023). This can be seen in Table 6.3 and Figure 6.2 below.

Table 6.3. Installed PV power of the three sub-markets in Sweden

Year	Off-Grid [MW]	On-Grid Distributed [MW]	On-Grid Centralised [MW]	Total [MW]
2020	15.38	1007.82	81.58	1104.78
2021	17.10	1453.33	133.84	1604.27
2022	18.34	2203.59	171.05	2392.98
2023	19.60	3736.90	238.68	3995.18

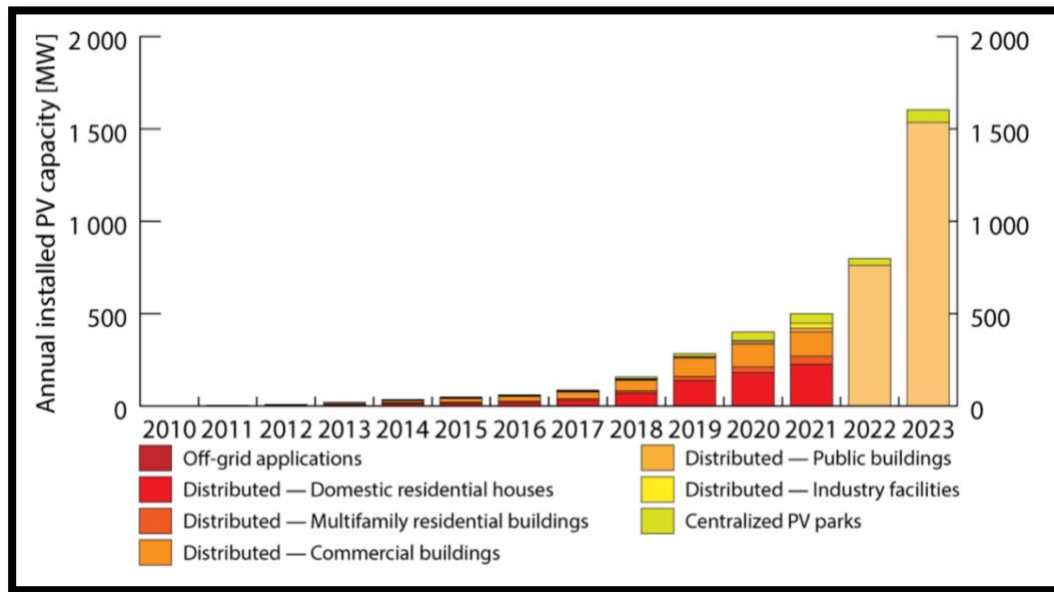


Figure 6.2. Continuous increase in PV installation over the years in Sweden (Westerberg, 2023)

From 2023, the Swedish PV installers and retailers have mentioned decrease in PV modules price to its end customers. Mentioning the last four years of data of PV module pricing, which indicates the PV market growth (Table 6.4).

Table 6.4. Typical module prices in Sweden over the years (Westerberg, 2023)

	Lowest price of a standard module crystalline silicon [SEK/Wp]	Highest price of a standard module crystalline silicon [SEK/Wp]	Typical price of a standard module crystalline silicon [SEK/Wp]
2020	2.5	6.6	4.0
2021	3.5	7.0	4.6
2022	2.6	7.8	5.6
2023	1.7	5.6	3.1

The overall cost of PV system has shown a decrease in 2023, making a more affordable renewable solution (Figure 6.3).

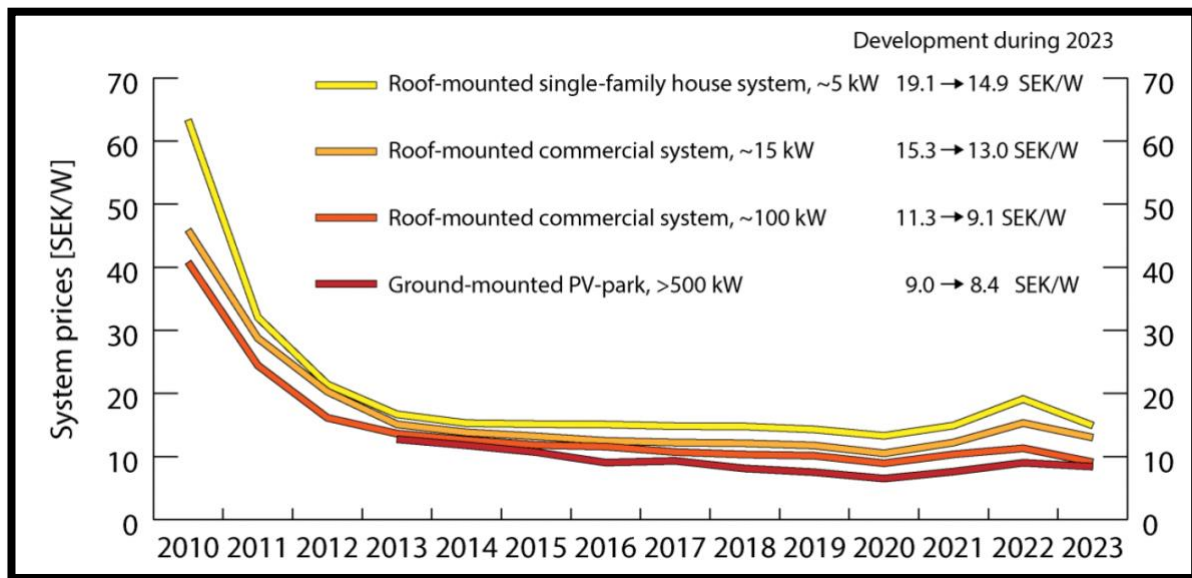


Figure 6.3. PV system prices (excluding VAT) in SEK/Watt (Westerberg, 2023)

The valuable capital subsidy program was terminated in 2022, which used to provide valuable data about PV system prices in Sweden. Currently to do the cost analysis, the data of 2023 from sales statistics have been shown in the report. The average loan rate in 2023 has been 4,8% for residential PV and 6,5% for commercial and ground mounted systems. Vattenfall in partnership with Handelsbanken provides solar loan at an interest rate of 6,06% which seems to be very competitive. Self-consumption of PV electricity is allowed in Sweden which seems to be a good business model driving the PV market. Since July 1st, 2021, energy tax on self-consumption has been increased from 255 KWp to 500 KWp, if the system size is less than 500 KWp, the solar producer does not need to pay any energy tax for self-consumed electricity (Westerberg, 2023).

From the above information, it is evident that the price of solar PV systems has been decreasing, and it is becoming more affordable over the years. There has been data from solar installers and retailers regarding the decrease in price of PV modules which has led to the upward trend of PV installations in Sweden. All these details are in favour of implementing solar carports in Huseby Bruk.

There are, however, a few concerns to consider such as weather impacts and energy storage. Snow accumulation on PV modules requires cleaning throughout the winter season which increases maintenance costs. Cloud cover also decreases PV yield during the autumn season. A PV system connected to the grid will not be able to store any amount of excess energy unless it is connected to battery storage or some sort of storage system. For these reasons, a solar carport together with biogas solution will prove beneficial for Huseby Bruk to meet the energy demand by means of sustainable energy. The implementation of these energy sources will not only solve the sustainable production of electrical energy but can also be an educational resource for visitors to Huseby Bruk to learn about the importance of renewable energy and its role in the energy transition.

6.2 Bioenergy

In relation to the objectives established at the outset of the project, the biogas solution satisfies most of the key requirements. It makes use of locally produced manure to generate renewable electricity, which contributes to a more sustainable, efficient, and resilient energy system. By supplying Huseby Bruk with locally sourced biogas, the generators can ensure continuous grid stability. Their ability to ramp up production quickly supports self-sufficiency, even during sudden surges in electricity demand or drops in production from intermittent renewable sources such as hydro and solar. An additional advantage is that biogas can be stored, making it a reliable option for balancing seasonal fluctuations in energy consumption. (European Biogas Association, 2023) This is particularly relevant for Huseby Bruk, where electricity demand is highest during the winter months due to significant heating needs. Biogas can be produced continuously throughout the year and stored for use during peak consumption periods, enhancing Huseby Bruk's overall energy security. Furthermore, an additional benefit of using an anaerobic digester is its byproduct, which can be used as organic fertilizer for nearby farms. This returns the favour to the farmers providing cow manure and further strengthening Huseby Bruk from an economic perspective (Kirk et al., 2019).

Despite these benefits, the main limitation of the biogas solution lies in its high capital and operational costs. To utilize local cattle manure, an anaerobic digester must be installed to convert the manure into biogas. This represents a substantial upfront investment. Additionally, a generator is required to convert the biogas into electricity, further increasing the overall system cost. Beyond the initial investment, the operation of such a system demands regular supervision. An on-site employee would be necessary to monitor the process and ensure the system operates reliably. Ideally, this person would live near Huseby Bruk and be available to respond to emergencies. However, given the area's demographics – predominantly elderly residents – it may be challenging to find local personnel with the appropriate technical expertise. Therefore, a trained operator may need to be sourced from a nearby town, adding to ongoing costs and potentially reducing response time in the case of system failures.

Another consideration is the noise associated with electricity generation from biogas. Unlike solar, biogas generators can produce noticeable levels of noise. To mitigate this, the plant would need to be strategically located to minimize disruption to the local community (Earth Care Equipments Pvt. Ltd, 2025).

Furthermore, since the electricity is generated using conventional generators rather than a CHP system, a significant portion of the energy—namely the heat—is lost. While a CHP setup could improve overall efficiency by capturing and reusing this waste heat for district heating, such a solution is not feasible under current conditions in Huseby Bruk. The town currently relies on electric resistor heating, and transitioning to a water-based district heating system would require major infrastructural changes (Sweco, 2025). These would include the installation of underground piping and modifications to individual buildings. Not only this would represent a considerable additional investment, but it would also be disruptive to residents and potentially impossible in the protected historic buildings. Given that the existing heating system functions adequately, it is unlikely that the community would welcome such intrusive modifications. (Energypedia, n.d.)

Biogas combustion is still considered a green energy source even if it produces CO₂ into the atmosphere. The basis of this classification is the fuel's source as well as its part in the short-term carbon cycle. Biogas comes from recently living organic materials, unlike fossil fuels which emit carbon that has been buried underground for millions of years. These comprise organic waste

including food scraps, animal manure, sewage sludge, and agricultural leftovers. Originally taken from the atmosphere by plants through photosynthesis, the carbon in these products then found its way into the food chain. The CO₂ emitted by processing biomass into biogas and burning simply returns to the atmosphere, so closing the carbon cycle (IEA Bioenergy, 2018).

Using organic waste that would otherwise break down naturally and immediately release methane (CH₄) straight into the atmosphere is another crucial feature of biogas generation. Over a 100-year period, methane is a strong greenhouse gas having a global warming potential 28-36 times that of CO₂. Biogas systems greatly lower the net greenhouse gas emissions by capturing this methane from anaerobic digestion and running it as fuel rather than by allowing organic matter to break down uncontrolled (International Energy Agency, 2021).

Furthermore, biogas is seen as renewable since its feedstock (organic waste) is always produced as a side effect of human and animal activities. There will thus always be a supply of biodegradable material fit for biogas generation if agricultural methods, food production, and wastewater treatment go on. This is not the case with fossil fuels, which are limited and cannot be replaced over human timeline.

Although burning biogas does produce emissions, its environmental effect overall is far less than that of fossil fuels. Furthermore, minimising their carbon footprint, modern biogas systems can be maximised for higher efficiency and even merged with technologies like carbon capture. Particularly as they replace fossil fuels and reduce methane emissions from traditional waste disposal techniques, biogas systems occasionally can even attain carbon-neutral or carbon-negative status.

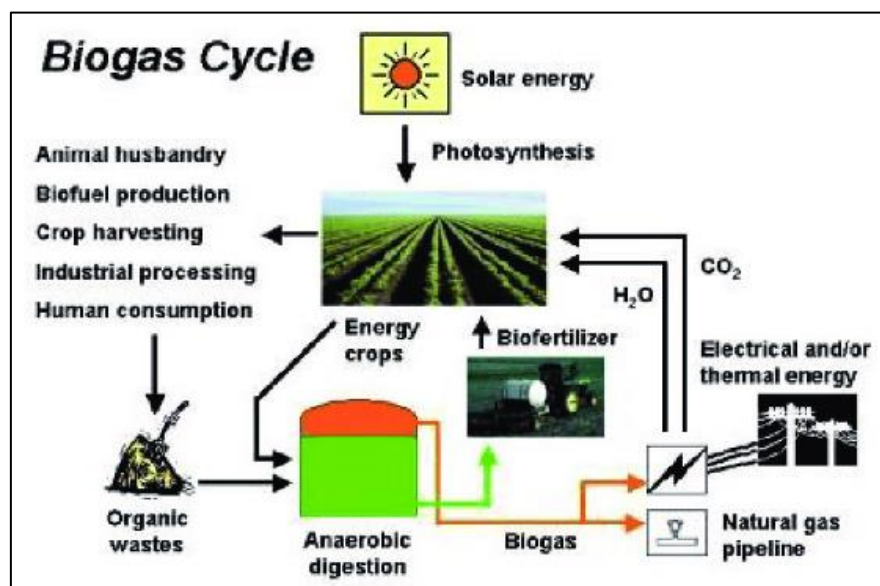


Figure 6.4. Biogas cycle regarding carbon emissions (Damyanova, S., 2020)

It is, however, important to note that biogas does generate emissions, even if it is regarded as a renewable energy source of power. Though these emissions do not always count in the long run, and often are not considered, there are not many people living in Huseby Bruk, hence these emissions could result in significant net carbon footprint per person. Indeed, utilising this solution would result in a far larger carbon footprint than a substitute solar powered one.

Nonetheless, when evaluated holistically—considering local resource availability, seasonal demand, storage capability, and emissions reduction potential—biogas remains a compelling option for Huseby Bruk’s transition toward renewable, self-sufficient energy.

6.3 Battery Energy Storage

By storing excess energy from renewable sources, including solar PV and hydropower, battery energy storage helps address the intermittency of generation, supporting the aim to create a sustainable and self-sufficient energy system.

Battery storage offers several key benefits that align with these objectives. Firstly, it enables peak shaving by storing energy during low-demand periods, such as midday solar surplus, and discharging it during peak times when electricity price is high, thus leading to potential cost savings (EVESCO, n.d.). Additionally, batteries enhance energy resilience by providing backup power during grid disruptions (AREVON, 2025), ensuring a more reliable energy system for residents and visitors. Secondly, with the integration of batteries, self-sufficiency is increased by storing surplus renewable energy for use during low-production periods and reducing reliance on the grids.

However, battery storage faces several disadvantages that challenge its feasibility. As mentioned before, the daily design battery can only provide limited energy, which is insufficient to meet the property’s annual demand, while its LCOS exceeds Sweden’s electricity tariffs. This diminishes the economic viability of battery storage. Moreover, daily storage designs cannot address the central issue of seasonal mismatches between generation and consumption—such as high summer solar production versus high winter heating demand—which is the primary purpose of implementing energy storage at Huseby Bruk. Therefore, given the limitations of battery storage, although it does not appear to be a cost-effective solution to achieve the project goal, it still offers beneficial contributions that can complement other renewable generations.

6.4 Integrated Solution

A combined solution merging solar and wind installations with biogas production (and an improvement of the hydro plant) would enable Huseby Bruk to become energy self-sufficient year around. This is the optimal solution for Huseby Bruk regarding sustainability and resilience. A possible combined solution is given in Figure 6.5 where the biogas was modelled to match the missing production

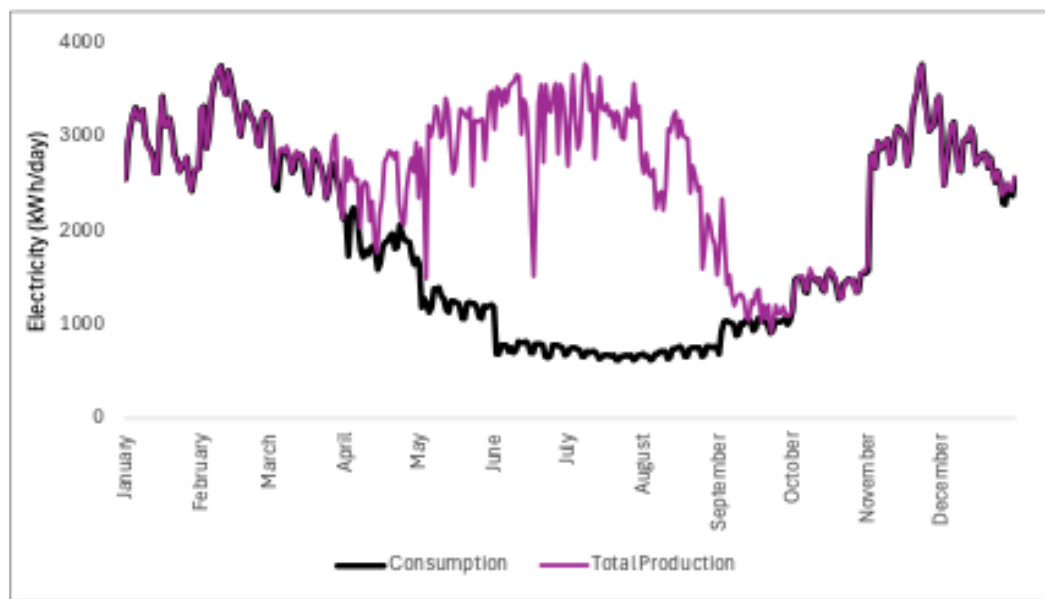


Figure 6.5. Combined solution where biogas production is modelled to meet demand

However, after reviewing final calculations and results, it was surprising to realize that the potential energy generated by biogas is much higher than expected. As a result, the relative contribution of solar to overall electricity generation appears limited. Nonetheless, this installation still plays an important role within the energy strategy of Huseby Bruk, particularly in demonstrating renewable technologies and supporting educational initiatives. It is important to note that while the solar plant does indeed produce significant amounts of electricity, most of it is produced during the summer, when the hydro plant produces most electricity, and there is no additional electricity needed.

Since the working methodology was carried out with the aim of favouring higher overall electricity production and energy resilience, it was decided to merge all available energy generation options. Indeed, only installing a solar plant would not allow Huseby Bruk to become energy self-sufficient. The same way, depending exclusively on biogas would reduce the diversity of the energy mix and put pressure on the farm owner, an independent stakeholder, all while not utilizing all available resources.

To be able to make effective use of all these proposed new renewable energies generated on the property it is important that Huseby Bruk first ensures that the electricity produced can also be directly used on the property to become truly energy independent and reduce electricity costs.

6.5 Future Work

Future work on this project should include a more detailed engineering analysis of combining the solar carport and biogas solutions and aligning energy production with Huseby Bruk's consumption needs. Although initially deemed economically inviable, changing the heating system from an electric heating system to, for example, a water-based heating system could increase the efficiency of energy used for heating the property significantly. A better assessment could be made through an in-depth analysis of this scenario. Finally, further investigation into the potential for a virtual grid at Huseby Bruk is recommended, as it is still a very promising solution for the high electricity cost in the property and is even more interesting if more renewable energy is added to the system.

7 Conclusions

Huseby Bruk is a historic property in southern Sweden that welcomes tourists to explore what life was like over a hundred years ago. The goal of this project is to continue Huseby Bruk's legacy of energy innovation and independence by increasing resiliency, decreasing energy costs, and attracting visitors to Huseby Bruk.

A virtual grid was determined to be a promising solution to lower grid connection costs, however, further investigation into virtual grid design is outside of the scope of this project as further work in this direction would be based more in regulations and legalities than in engineering. Nevertheless, although this was a big boundary for the skill set of the team, it still looks to be a promising solution for Huseby Bruk in general, especially considering that since they will again be one of the first properties in Sweden to implement this, enhancing their legacy of being at the forefront of the electricity development.

The implementation of the solar carports will not only generate electrical energy from renewable source but will also be beneficial in providing a proper shading arrangement for the vehicles in the Huseby Bruk's parking lots. The output from the solar carports will approximately meet most of the yearly energy demand of Huseby Bruk, but, considering the sunlight dependency, the output from the solar carports will not be even throughout the year. Huseby Bruk will not be net-zero as per an hourly and daily basis without the implementation of storage or any other source of renewable energy. The positive impacts of the PV system in the solar carport will be reducing carbon footprint, lowering the energy price once the payback period is over, and increasing the energy independence of Huseby Bruk. However, due to climate dependency, the solar carport system will need battery storage to store the surplus solar energy or can be summed up with another source of renewable energy to meet the total energy demand of Huseby Bruk throughout the year on an hourly basis.

Biogas production through an anaerobic digester has great potential to solve the intermittency problems Huseby Bruk currently faces in their electricity profile, as biogas can be produced when the feedstock is available and used to produce electricity on demand when needed. Storing the biogas when there is overproduction and generating electricity when there is high demand would allow Huseby Bruk to become more energy independent, as well as generating fertilizer as a byproduct of the anaerobic digestion. However, the cost of electricity generation through biogas and the infrastructure needed are relatively expensive and not competitive with other renewables or the grid electricity price. Furthermore, it is also of note that although bioenergy is considered a green energy source there is still CO₂ emitted while producing energy with biogas.

The most significant challenge towards fulfilling Huseby Bruk's goal of independence and resiliency is the current structure and management of the local electrical grid. Once an appropriate solution to the grid management problem has been found and the produced electricity can also be consumed locally, the solution proposed in this report should allow Huseby Bruk to become (almost) energy independent. With bioenergy as the backbone to solving the seasonal mismatch of the electricity pattern, Huseby Bruk, as an energy community with its own grid, would be able to again take up its role at the forefront of energy innovation.

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9 Appendices

Appendix A: Energy Network Structures & Regulations

The regulatory framework for electrical grids in Sweden has historically made it difficult for users to share energy between nearby electrical grid subscription points. Electricity users are required to sign an electricity network subscription per input or output point to the electricity grid in order to have the right to consume or produce electricity. All electricity must be transferred through the grid and cannot be directly transferred between subscription points. This has made it difficult for users to share electricity between subscription points and for prosumers – customers who both produce and consume energy – to directly consume the electricity they produce. For example, if one person in a neighbourhood has solar PV panels and produces more energy than they use, they cannot share that electricity directly with a neighbour through the grid. For Huseby Bruk, this means that electricity produced on one part of the property cannot be directly used on another part of the property; the part of the property producing electricity must sell it to the grid and the part consuming electricity must buy it from the grid. However, regulations have recently changed in Sweden, opening more possibilities for energy sharing.

Parallel Network

One potential solution considered by Sweco is for Huseby Bruk to build its own separate electrical grid, or parallel network. This is legally possible but logistically and financially difficult. Despite the legality of building a parallel network, Sweco previously determined that it would not be a desirable solution for multiple reasons. Firstly, it is not a sustainable solution. Secondly, it would require increased construction time and disruption to the physical property of Huseby Bruk. Finally, a parallel network would demand significant resources for construction, maintenance and management. In this scenario, Huseby Bruk would own their electrical grid. Therefore, if anything broke or needed maintenance, Huseby Bruk would be responsible for fixing it.

Virtual Grid

A second solution considered by Sweco is a virtual grid. A virtual grid is an integration of small-scale prosumers – individuals who both produce and consume energy – in order to operate as a single controlled entity (Sardi et al., 2021). The prosumers share the electricity produced by their own distributed energy resources and the economic profits through smart electricity meters in the existing grid (Fridstrand, 2024). The smart meters provide real-time data on electricity production and consumption, allowing virtual grids to match local demand with production and supply from the grid. Therefore, power system efficiency can be enhanced by utilizing virtual grids (Frieden et al., 2021).

Virtual grids enable residential prosumers to actively participate in the electricity market. Prosumers can share energy with neighbours during peak demand periods when grid electricity is expensive and sell excess energy back to the grid at high prices, thus creating economic benefits for the prosumer. By facilitating local energy sharing, reducing reliance on centralized power, and optimizing market participation, virtual grids contribute to a more sustainable and cost-effective energy system. These advantages position virtual grids as a transformative solution for modern energy challenges.

Historically, virtual grids were not possible in Sweden due to energy sharing regulations that required all energy produced to be sold to the grid. Regulations are changing, however, and virtual grids are gaining popularity. New EU rules encourage the sharing of local energy through so called

“energy communities” to support increased production of sustainable energy. The vision is for energy communities to be a way to restructure current energy systems by empowering citizens to drive the energy transition locally and benefit from better energy efficiency, lower bills, reduced energy poverty, and more local green job opportunities (European Commission, n.d.). Despite this momentum, virtual grids remain largely untested and may face additional regulatory hurdles or other barriers to implementation.

There are not many examples of virtual grids in Sweden, as regulations have only recently changed. However, one case study was identified. A company called sonnen debuted a virtual power plant (sonnenVPP) in Sweden in March 2024. This company sells intelligent home energy storage systems called sonnenBatteries. The sonnenVPP enables households to use their existing solar PV systems and sonnenBatteries for self-consumption of clean energy at home. Simultaneously, the storage system is interconnected with other residential storage systems to form a large virtual battery. That battery network stands ready to stabilize the grid in the event of grid fluctuations that could cause a disruption in the power supply to homes across Sweden (sonnen, 2024). Sonnen operates in a variety of other countries but only recently got approval from the Svenska Kraftnät, the Swedish transmission system operator (TSO), to operate in Sweden.

Sweco has determined that a virtual grid is a promising solution at Huseby Bruk, and that further research is warranted to explore this potential. A virtual grid could increase incentives to share energy, benefit future energy systems in Sweden and beyond by setting a precedent, and lower electricity costs for the property. A virtual grid would allow Huseby Bruk to directly use the energy produced on the property. An additional economic benefit of a virtual grid is significantly decreased grid subscription fees, as the resulting system would only have one grid subscription point. With a virtual grid, Huseby Bruk would be able to net meter over their property by using one a set of smart meters resulting in one electrical bill with lower costs instead of many. Huseby Bruk would still be connected to the regional electrical grid and any additional surplus electricity produced could still be sold back to the grid for profit. Unlike a parallel network, a virtual grid can be implemented without the need to invest in or manage any additional electricity networks as well as that the DSO will still be responsible for the grid. The National Property Board is currently in discussions with the local grid company on how it can increase the local usage of renewables on the property at Huseby Bruk.

Appendix B: Wind Analysis

Firstly, it was just considered the installation of the WS-015 and WS-030 of WindSite company. It has been taken WindSite as a reference retailer because of the transparency with which it provides precise data about its turbines' performances, aspect that is rarely clear for other retailers who require to be further contacted to collect precise data.

The WS-015 and WS-030 models have a height of 1.5 meters maximum (WindSite, 2020), which was acceptable for the property's restrictions. The WS-060 has a height of 2.4 meters (WindSite, 2020), thus resulting in disturbance in the landscape. Subsequent models were discarded because of an even greater height. Utilizing the parking lot has however shown the possibility of installing bigger turbines such as the WS – 2, which has a total height of around 2.8 meters, a diameter (D) of 1 meter and a power output between 0,2-1 MWh per year in rated conditions (WindSite, 2020). The output in Watt could be calculated considering an extremely optimistic capacity factor of 0,43 (Global Wind Atlas, 2024) at a height of 100 meters, which resulted in a power output of 53-265 W. It is an optimistic assumption because it is measured at 100 meters, while the installation would be at 10 meters.

Considering structures higher than 10 meters is believed to not bring relevant advantages: at a 50 meters height an average wind speed of 6.37 m/s is measured (Global Wind Atlas, 2024), only 1.88 m/s more than at 10 meters (Global Wind Atlas, 2024). Structures higher than 20 meters are even required to obtain a building permit, while structures with a maximum height of 20 meters and maximum rotor diameter of 3 meters may be exempted from a building permit if they are not located on a building or are not closer than their own height to a building (European Commission). Finally, considering all the possibilities, if installing micro-wind turbines, it would be preferable to choice for a 10 meters high configuration. Nevertheless, this choice needs to be evaluated considering the total costs, energy output at available wind speed conditions, energy usage (if utilized or sold to the grid) and possible educational/attractive purposes.

Each of the VAWTs should occupy an area of 36 m² (turbines will be spaced at 6D meters x 6D meters X and Y distance), to lower possible turbulences and optimize the performance (Alqahtani et al, 2021). In this case, it would be possible to install a maximum turbine count of $3550 \div 36 = \sim 98$ turbines.

Approximate Annual Power Production at Annual Average Wind Speeds*						
m/s	Output per Model (kWh/year)					
	WS-015	WS-030	WS-060	WS-2	WS-4	WS-12
4	15	30	60	120	400	720
5	40	80	160	700	2000	8640
8	96	192	384	1800	5200	22464
10	206	413	826	3900	11100	48298
*Indicative values only. Actual power output will vary according to conditions.						

Figure B1. WindSite output per model (WindSite)

Considering a yearly average wind speed between 4 and 5 m/s, a total of 98 WS-2 turbines would generate an estimated total output of:

- m/s: $120 \text{ kWh} \times 98 \text{ turbines} = 11,760 \text{ kWh/year}$

- m/s: $700 \text{ kWh} \times 98 \text{ turbines} = 68,600 \text{ kWh/year}$

An economic plan should consider the turbine hardware, installation, permitting and maintenance costs. The following data can be taken as an estimation (Project DrawDown). Detailed calculations have to consider a precise model of turbine from a precise retailer. It is difficult to find detailed information about the price of a product without contacting and meeting the retailer.

- Installation costs: \$4,536–6,481 per kW (€4,358 to €6,229 per kW)
- Permitting costs: to be defined by the county administrative board (European Commission).
- Maintenance and other fixed costs: 0.03€/kWh for variable operation and maintenance costs and 121,76€/kW for fixed costs

For example, installing a VAWT WS-2 turbines from WindSite, would cost the following estimated prices per turbine:

- Installation Cost:
 - $0,053 \text{ kW/turbine} \times 4,358 \text{ €/kW} = 231 \text{ €}$ (lowest forecasted price)
 - $0,265 \text{ kW/turbine} \times 6,229 \text{ €/kW} = 1,650 \text{ €}$ (highest forecasted price)
- Permitting costs: to be defined by the county administrative board
- Fixed costs:
 - $200 \text{ kWh/turbine} \times 121.76 \text{ €/kWh} = 24,351 \text{ €}$ (lowest forecasted price)
 - $1000 \text{ kWh/turbine} \times 121.76 \text{ €/kWh} = 121,760 \text{ €}$ (highest forecasted price)
- OPEX costs:
 - $200 \text{ kWh/turbine} \times 0.03 \text{ €/kWh} = 6 \text{ €}$ (lowest forecasted price)
 - $1000 \text{ kWh/turbine} \times 121.76 \text{ €/kWh} = 30 \text{ €}$ (highest forecasted price)

A return of investment (ROI) can't be calculated in the case of focusing the project into educational purposes, since not enough data are available.

Appendix C: Solar PV Supplementary Information

Solar PV Cell Technology

Solar PV technology converts sunlight directly to electricity. Solar cells composed of p-type and n-type semiconductors, joined together to form p-n junction where electric field is formed due to flow of electrons to positive p-side and holes to negative n-side. Sunlight is composed of photons which are small bundles of electromagnetic energy. When light of acceptable wavelength falls incident on the solar PV cell, the photon energy is transferred to an atom of the semiconductor p-n junction. The energy transferred to the electrons causes them to jump to the higher energy band named the “conduction band” leaving a hole in the “valence band”. This electron movement due to added energy creates electron-hole pair. The electric field causes the electron and the hole to move in opposite directions where the motion of the electrons to the n-side and the holes to the p-side gives rise to electric current in the solar cell (Energy Education, n.d.).

Different Types of PV Cells

There are three main types of PV cells that are dominant in the market: monocrystalline silicon, polycrystalline silicon and thin film. The monocrystalline cells use pure silicon, which is polished and doped to create a p-n junction. An anti-reflective coating is given to the top and the rear part and then wired with other cells to form a solar panel. These cells are highly efficient and require a labour intensive and time-consuming process making it more costly compared to polycrystalline and thin-film cells. Polycrystalline cells do not use single uniform crystal of silicon but use small grains of the crystals which reduces the efficiency of the cells but is more cost effective compared to monocrystalline cells. Polycrystalline cells mostly constitute the global PV market (Precedence Energy and Power Experts, 2023). Thin film cells compared to crystalline cells are less dominant in the PV industry, but they are much more flexible and durable. They are generally made of very less silicon deposit on a glass substrate, due to less usage of raw material, thin film cells are much cheaper compared to crystalline cells. The efficiency of thin film cells is greatly reduced because the silicon is much less ordered as compared to crystalline cells. Monocrystalline solar PV modules have been considered for Huseby Bruk calculations.

Solar Pricing Information

The solar pricing market in Sweden has been volatile due to constant improvement in solar technologies and advancements in the solar cell technologies. Table C1 below refers to the best of knowledge solar system prices considering the type of installations, system sizes and grid connectivity.

Table C1.1 Estimated PV system prices in 2023 based on sales report (Westerberg, 2023)

Category/Size	Typical applications and brief details	Current prices [SEK/W _p]
Off-grid 2 kW	A stand-alone PV system is a system that is installed to generate electricity to a device or a household that is not connected to the public grid. The price is for a small off-grid system on a cottage for seasonal use (summer) that is not connected to main grid.	20–27
Residential BAPV 5-10 kW	Grid-connected, roof-mounted, distributed PV systems installed to produce electricity to grid-connected households. Typically roof-mounted systems on villas and single-family homes.	11–19
Small commercial BAPV 10-100 kW	Grid-connected, roof-mounted, distributed PV systems installed to produce electricity to grid-connected commercial buildings, such as public buildings, multi-family houses, agriculture barns, grocery stores etc.	7–16
Large commercial BAPV 100-250 kW	Grid-connected, roof-mounted, distributed PV systems installed to produce electricity to grid-connected large commercial buildings, such as public buildings, multi-family houses, agriculture barns, grocery stores etc.	6–11
Industrial BAPV >250 kW	Grid-connected, roof-mounted, distributed PV systems installed to produce electricity to grid-connected industrial buildings, warehouses, etc.	5–11
Small centralised PV 1-20 MW	Grid-connected, ground-mounted, centralised PV systems that work as central power stations. The electricity generated in this type of facility is not tied to a specific customer and the purpose is to produce electricity for sale.	5–9
Large centralised PV >20 MW	Grid-connected, ground-mounted, centralised PV systems that work as central power station. The electricity generated in this type of facility is not tied to a specific customer and the purpose is to produce electricity for sale.	not applicable

Table C2. Design comparisons between single-sloped and dual-sloped carports

	Single-sloped Solar Carport	Y-shaped (Dual-sloped) Solar Carport
Area	5144 m ²	5144 m ²
Slope	6 degrees	6 degrees
Azimuth	260 °	260 ° 80 °
Front post	2.286 m	2.286 m
Back post	3.048 m	3.048 m
Distance in between carports	7 m	7 m

The PVWatts software shows the output from the two different designs of carports. Both the designs have a yearly generation of approximately 252 MWh/year (Tables C3 and C4).

Table C3. PVWatts showing yearly energy production for the single sloped solar carports

RESULTS		
251,706 kWh/Year*		
 Print Results		
Month	Solar Radiation (kWh / m ² / day)	AC Energy (kWh)
January	0.25	1,872
February	0.83	6,640
March	2.29	21,372
April	3.82	33,569
May	4.52	39,915
June	5.02	42,620
July	5.26	45,556
August	3.45	30,395
September	2.26	19,546
October	0.98	8,501
November	0.16	1,037
December	0.12	683
Annual	2.41	251,706

Table C4: PVWatts showing yearly energy production for the Y-shaped (dual sloped) solar carports with total of 251883 KWh/year

RESULTS			126,030 kWh/Year*		
Print Results			Print Results		
Month	Solar Radiation (kWh / m ² / day)	AC Energy (kWh)	Month	Solar Radiation (kWh / m ² / day)	AC Energy (kWh)
January	0.24	892	January	0.25	936
February	0.81	3,239	February	0.83	3,320
March	2.25	10,509	March	2.29	10,686
April	3.84	16,965	April	3.82	16,784
May	4.43	19,611	May	4.52	19,958
June	5.04	21,475	June	5.02	21,310
July	5.28	22,944	July	5.26	22,778
August	3.49	15,387	August	3.45	15,198
September	2.29	9,905	September	2.26	9,773
October	0.97	4,257	October	0.98	4,250
November	0.16	509	November	0.16	518
December	0.12	338	December	0.12	341
Annual	2.41	126,031	Annual	2.41	125,852

Detailed Cost Benefit Analysis

The detailed cost benefit analysis has been done considering the system size of 383 kW and production of 252 MWh/year. The best-case and the worst-case scenario for CAPEX and OPEX have been calculated along with annual revenue for the first year and the total revenue considering degradation rate of 0.05% and electricity sale price 0.2 Euro/KWh.

Sweden's Green Tax incentive has also been considered at 19.4% and calculated the best-case and the worst-case CAPEX calculation and OPEX calculations have been done over the period of 25 years and calculated the net profit over 25 years for both the cases.

Payback period, IRR, and Net Present Value (NPV) with 5% discount rate has been calculated for both the cases. All the detailed calculations are shown in Table C5 and Table C6, respectively.

Table C5. Calculation of CAPEX, OPEX and Total Revenue

System Size			383 kW	
Yearly Energy Production			251884 kWh/year from PV Watts	
Daily energy Production			690.09 kWh/day	
Considering 5 hour of energy production				
Hourly Energy Production			138.02 kWh/hour	
Hourly Energy Production (each carport)			69.01 kWh/hour	
Detailed CAPEX				
Component	Cost per kW (EUR) min	Cost per kW (EUR) max	Total Cost (EUR) min	Total Cost (EUR) max
Solar Panels	250	350	95750	134050
Inverters	120	180	45960	68940
Mounting structure and carport frame	350	500	134050	191500
Electrical Wiring & BOS	80	120	30640	45960
Labor and Installation	200	300	76600	114900
Grid connection and permitting	50	100	19150	38300
Misc Costs (Design, Engineering, Prep)	50	100	19150	38300
			EUR 4,21,300.00	EUR 6,31,950.00
Detailed OPEX				
Expense Category	Cost per kW/year (EUR) min	Cost per kW/year (EUR) max	Total cost (EUR/year) min	Total cost (EUR/year) max
Maintenance and cleaning	10	20	3830	7660
Inverter replacement	8	12	3064	4596
Insurance and misc.	5	10	1915	3830
			EUR 8,809.00	EUR 16,086.00
Energy Production & Revenue				
Electricity Sale Price	0.2 Euro/kwh			
Annual revenue (year1)	EUR 50,376.80			
Degradation Rate	0.005			
Number of years	25 years			
Total Revenue	EUR 11,86,673.45			

Table C6. Calculation of payback period, IRR and NPV

Tax Incentive (Grön Teknik Deduction)		
Swedish Green Tech Deduction	0.194	
Min. CAPEX after Initiative	EUR 3,39,568.00	
Max. CAPEX after Initiative	EUR 5,09,352.00	
Payback, ROI, IRR, and NPV Calculation		
Total OPEX over 25 years (Min)	EUR 2,20,225.00	
Total OPEX over 25 years (Max)	EUR 4,02,150.00	
Net profit (Min Scenario) over 25 years	EUR 6,26,881.00	
Net profit (Max Scenario) over 25 years	EUR 2,75,172.00	
Simple Payback period		
Min. Case	7 years	
Max. Case	10 years	
IRR Calculation		
Min Case	10.83 %	
Max Case	3.78 %	
NPV Calculation (at 5% discount rate)		
NPV Min. Case	EUR 2,13,599.88	In 25 years
NPV Max. Case	EUR 3,865.81	In 35 Years

Appendix D: Energy Storage Calculations

Sizing Result of Energy Storage – Yearly, Monthly, and Weekly

Table D1 and Figures D1-D5 show the sizing result of yearly, monthly, and weekly designs for an energy storage system. Since yearly storage shows a relatively low utilization rate but requires the highest capacity, which could result in a significantly high investment cost, it is not considered as a feasible option. As for monthly and weekly design, they provide only 14.04% and 12.91% of the energy discharged by yearly design storage, respectively (Table D1), and cannot address the seasonal mismatch between summer generation and winter consumption, while still requiring significant capacity and investment costs. Therefore, monthly and weekly designs are not feasible either, as they do not provide a cost-effective solution.

Table D1. Energy storage sizing results

Time Scale	Capacity (kWh)	Total discharged Energy (kWh)	Discharged Energy compared to yearly (%)
Yearly	134299.92	173766.41	100
Monthly	981.90	24400.51	14.04
Weekly	861.28	22435.61	12.91
Daily	388.96	12845.81	7.39

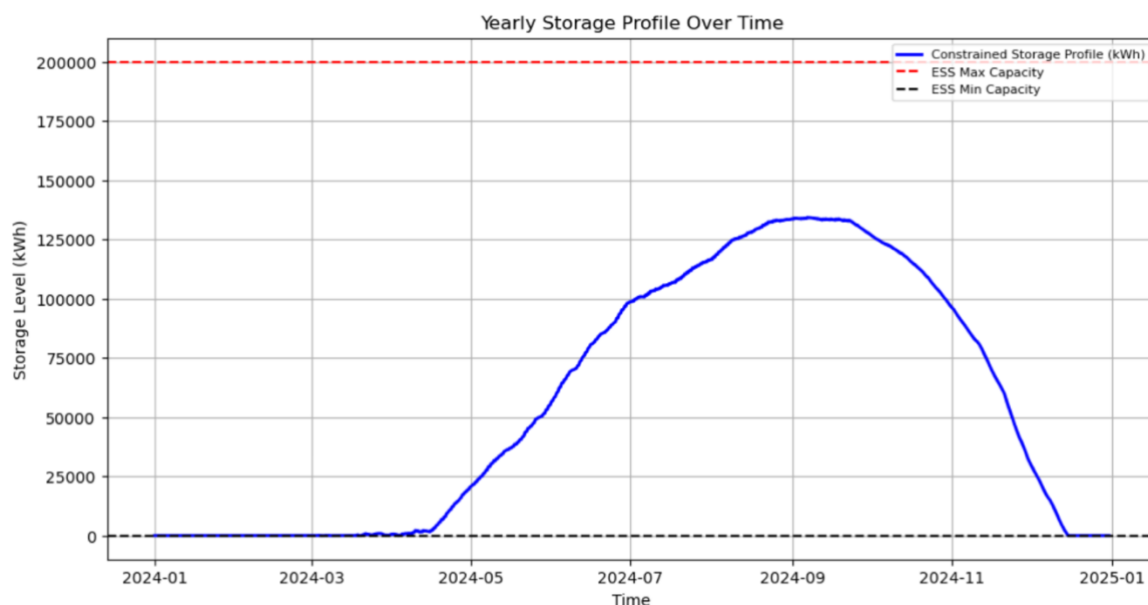


Figure D1. Yearly storage profile over time

Figures D2 and D3 show the monthly storage model results.

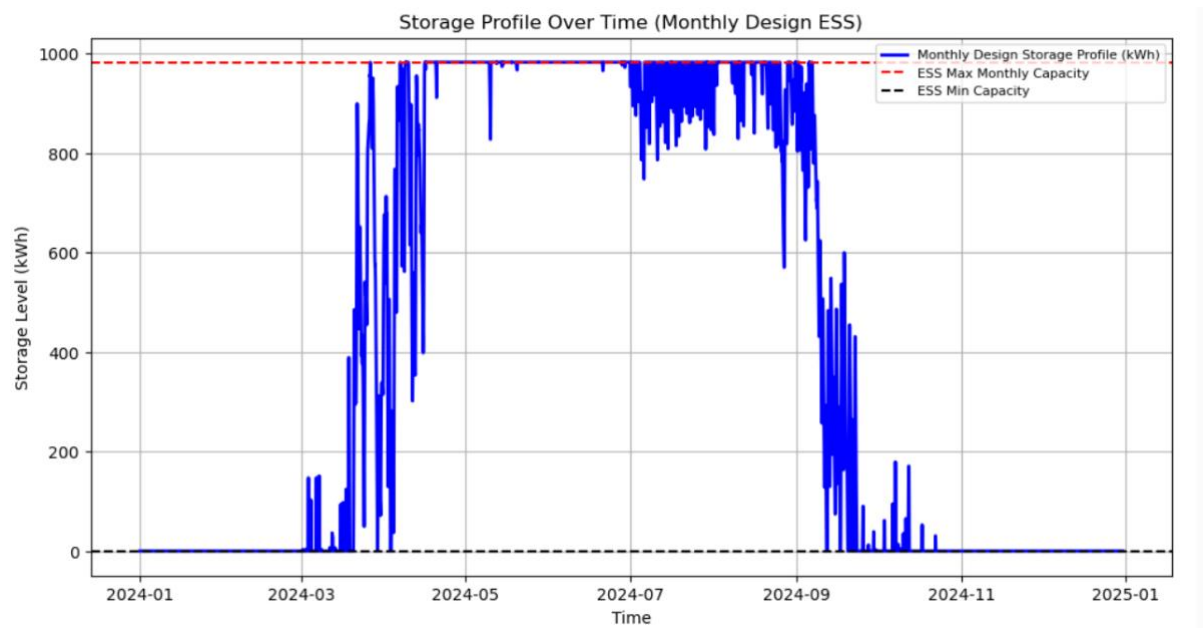


Figure D2. Monthly storage profile over time

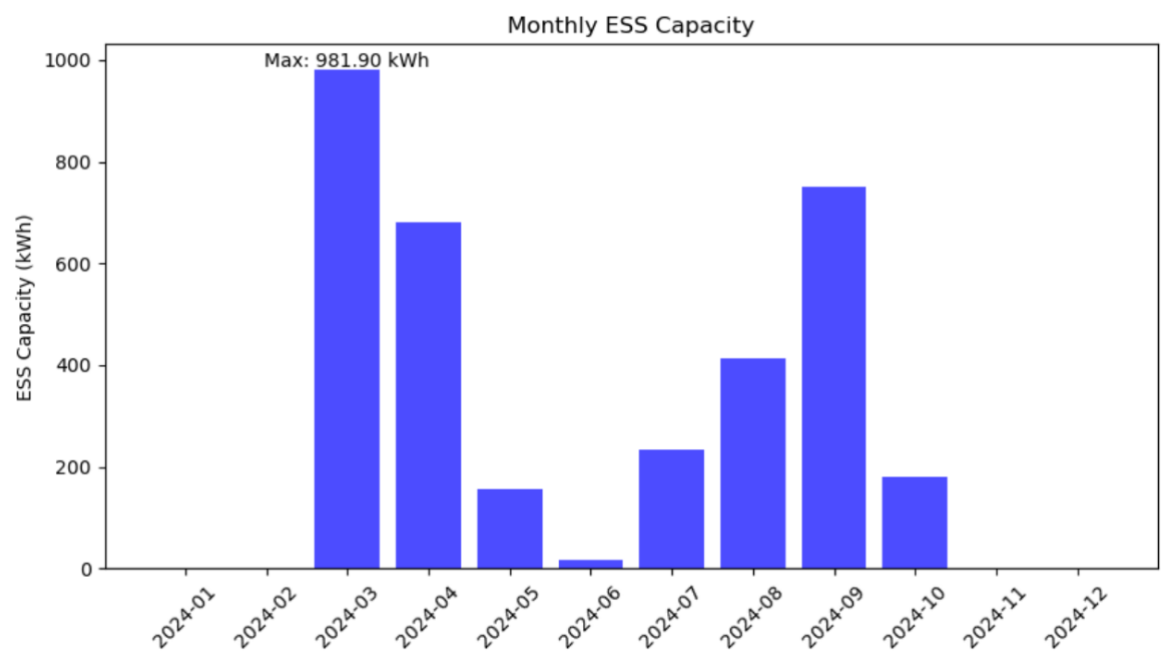


Figure D3. Monthly EES capacity

Figures D4 and D5 show the weekly storage model results.

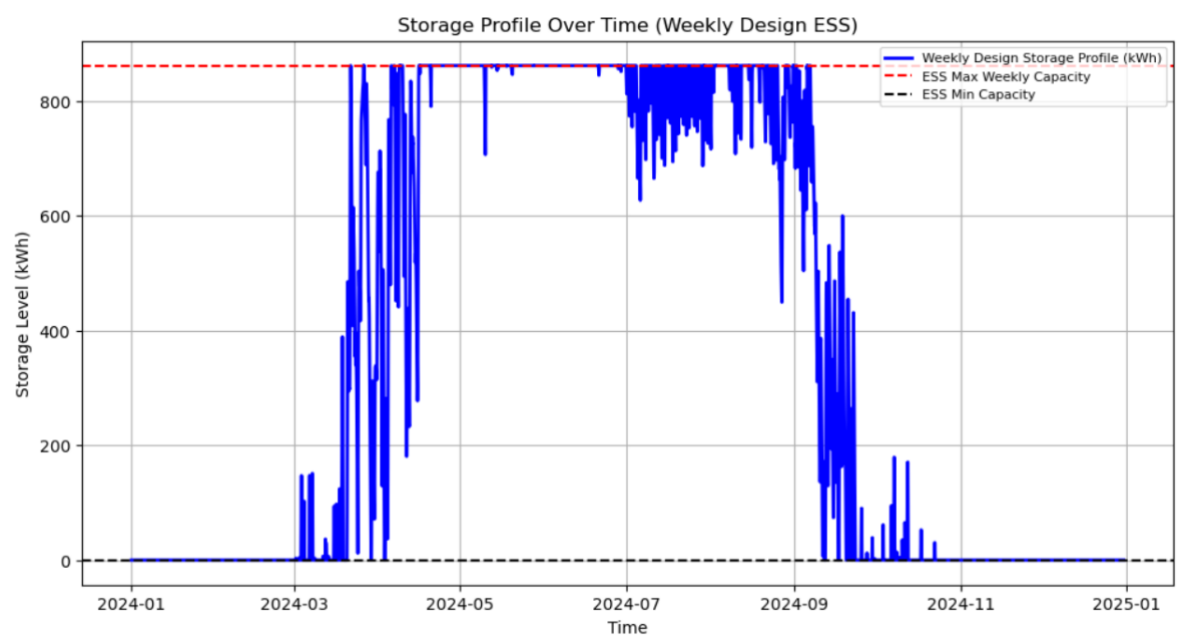


Figure D4. Weekly storage profile over time

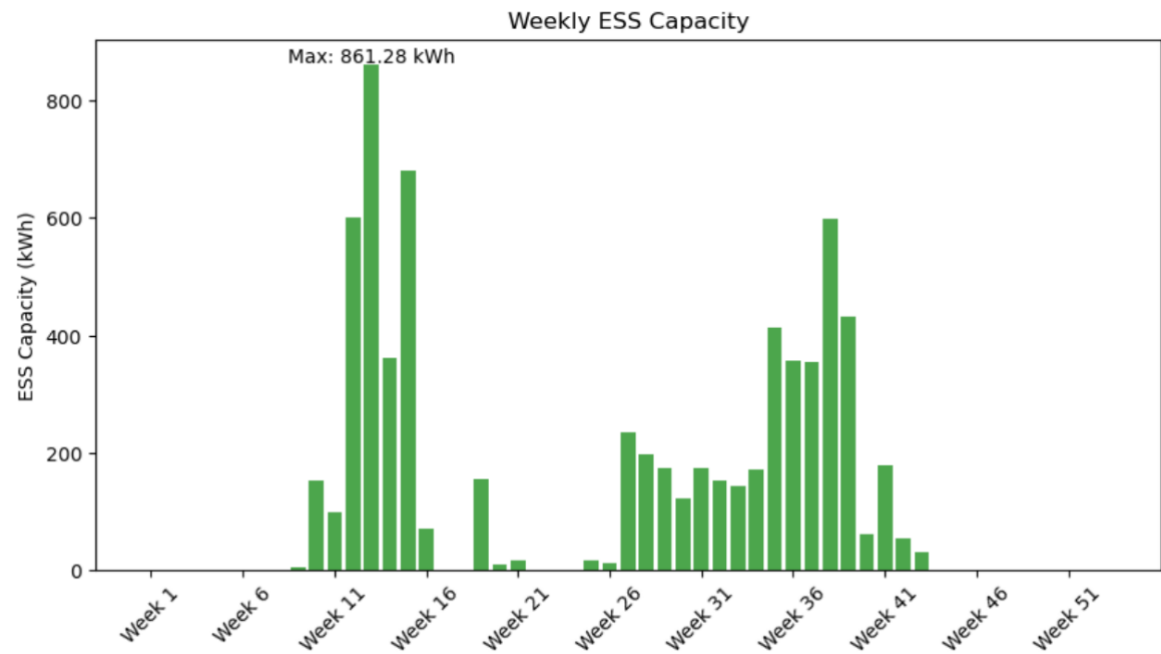


Figure D5. Weekly EES capacity